

Mountain Wellness & Adventure Retreats

Introduction to Computational Thinking and Data Science

Mountain Wellness Retreat Assessment

Client / Scenario: Mountain Wellness Adventure

Client: Mountain Wellness & Adventure Retreats
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Executive Summary

The client, Mountain Wellness & Adventure Retreats, must decide between Guarda and Vila Real for a premium mountain wellness resort in Portugal. The decision depends on more than average climate: a successful retreat needs comfortable outdoor days, reliable seasonal programming, manageable operating risks, and an atmosphere consistent with a high-end wellness experience.

This report analyses daily meteorological data from 1 January 2018 to early 2026 for both locations. The revised analysis makes factor priorities explicit, distinguishes annual comfort from seasonal business value, and connects each result directly to operational decision-making. A predictive component was also added to estimate whether a day is likely to be suitable for outdoor wellness activities using location, seasonality, and atmospheric indicators, while avoiding the direct rule variables that define an OK day.

The evidence supports a clear recommendation: the client should prioritise **Guarda** for the proposed premium mountain wellness retreat. The recommendation is not based on a large annual difference in total OK days, since Guarda and Vila Real are close overall (623 vs. 612). It is based on the stronger alignment between Guarda's climate profile and the business model. Guarda provides substantially more OK days in summer (276 vs. 211), fewer hot days (159 vs. 270), fewer warm nights (15 vs. 75), fewer rain days (856 vs. 1,039), lower OK-day variability across years (13.5% vs. 18.0%), and a brighter atmospheric profile.

Vila Real remains a credible alternative if the client decides to prioritise spring programming or winter mildness. It has fewer freezing days and more spring OK days. However, these advantages are less decisive for a resort concept centred on outdoor wellness, mountain atmosphere, and peak-season guest experience. Vila Real's disadvantages occur mainly in the periods and conditions that most directly affect revenue quality: summer heat, warm nights, rainfall disruption, and cloudier atmospheric conditions. Guarda therefore offers the stronger climatic platform, provided that the investment includes winter heating, insulation, and indoor wellness programming to manage its colder low season.

Chapter 1

Problem Framing and Context

1.1 Client Context and Objective

Wellness tourism has been increasing consistently in recent decades, driven by a stronger concern with health, mental well-being, and quality of life. Within this context, destinations that combine tranquillity, nature, and outdoor activities have become especially attractive. Mountain wellness retreats stand out because they can offer silence, fresh air, scenic views, and favourable conditions for restorative experiences such as yoga, meditation, walking, and seasonal wellness programmes.

However, the success of a mountain wellness retreat depends largely on the characteristics of its location. Among these factors, climatic conditions play a central role, since they directly influence guest comfort, the feasibility of outdoor activities, the perceived quality of the environment, and the operating costs of the hotel itself. A location that is too cold, too rainy, too cloudy, or too unstable across the years may reduce both the attractiveness of the experience and the long-term profitability of the investment.

This is especially relevant in the case of mountain wellness tourism. Unlike other tourism models that may rely more heavily on urban attractions or indoor infrastructure, this type of retreat depends strongly on environmental quality. Walks in nature, contemplative experiences, outdoor exercise, scenic relaxation, and the general feeling of calm all become more or less viable depending on temperature, rainfall, wind, and atmospheric clarity.

The client, *Mountain Wellness & Adventure Retreats*, is a developer of premium resorts seeking to expand its operations in Portugal. The business model is centred on high-end wellness experiences, including yoga retreats, meditation programmes, outdoor adventure trekking, and seasonal packages focused on rest and recovery. These activities are highly sensitive to meteorological conditions, which means climate is not a secondary contextual

feature but one of the main determinants of business viability.

The company is currently considering two possible investment alternatives in Portugal: Guarda and Vila Real. The objective of this report is to transform the available meteorological evidence into a structured investment recommendation. The analysis therefore evaluates which location offers the stronger climatic platform for a premium wellness retreat, considering outdoor usability, seasonal revenue potential, operating risk, atmospheric quality, and planning reliability.

1.2 Decision Alternatives

The decision consists of choosing between two inland mountainous locations in Portugal: Guarda and Vila Real. Both locations are recognised for cooler climates than coastal areas and, at first sight, both appear compatible with the strategic positioning of a mountain wellness retreat focused on tranquillity, fresh air, and nature-based experiences.

Even so, the two alternatives do not represent the same type of climatic opportunity. Guarda is generally associated with colder conditions, lower annual precipitation, and greater atmospheric clarity. Vila Real appears thermally milder during the colder part of the year, but also rainier and more exposed to summer heat. This means that the decision cannot be reduced to a simplistic comparison such as “which city is colder” or “which city is warmer”. The relevant issue is which climatic profile is more suitable for the specific type of premium retreat the client intends to develop.

In practical terms, the comparison must consider not only annual averages, but also the frequency of very cold days, the frequency of hot summer days and warm nights, the reliability of dry conditions for outdoor programming, and the consistency of the general atmospheric experience associated with the idea of a wellness retreat. Both locations may offer advantages, but they imply different forms of operational burden and different risks for guest experience.

For the purpose of this project, the “better” alternative is not simply the one with the mildest climate. It is the one whose climatic profile most effectively supports guest comfort, operational stability, predictable planning, and the consistent delivery of a premium wellness atmosphere.

1.3 Key Questions and Success Criteria

To support the client’s decision, the analysis is structured around four decision-oriented questions:

1. **Comfort:** Which location feels better for guests most of the time?
2. **Predictability:** Which location is more predictable and less variable across the years?
3. **Operations:** Which location is likely to cost more to operate or to be more seasonal?
4. **Atmospheric quality:** Which place offers the more restorative and visually appealing wellness environment?

These questions are useful because they translate a generic climate comparison into a business decision. Rather than asking only which place has “better weather”, the analysis asks which climatic profile better supports a premium retreat model.

The first question, comfort, concerns the conditions that guests will feel directly. A retreat should offer a large number of days in which temperatures are pleasant, rain does not interrupt activities, and wind does not reduce the usability of outdoor spaces. The second question, predictability, matters because the client is not only interested in average conditions, but in planning reliability. A location with acceptable means but strong year-to-year instability may be less attractive from an investment perspective than a location with slightly worse averages but greater regularity.

The third question concerns operations. Climatic conditions affect heating and cooling needs, maintenance requirements, scheduling constraints, and the risk of stronger or weaker low seasons. The fourth question concerns atmospheric quality and perceived wellness environment. A wellness retreat depends not only on physical comfort, but also on brightness, openness, scenic quality, and the psychological effect of clear skies and pleasant environmental conditions.

Chapter 2

Data Understanding

2.1 Data Sources

The analysis uses historical daily meteorological data obtained through the Open-Meteo archive API for the two candidate locations, Guarda and Vila Real. The datasets cover the period from 1 January 2018 to early 2026, which provides a sufficiently long horizon to compare average conditions, seasonal behaviour, and inter-annual variability.

According to the technical documentation supplied for the project, the Open-Meteo Archive API was used at daily resolution. This level of aggregation is appropriate for structured meteorological comparison because it preserves enough detail for frequency counts, seasonal analysis, and descriptive statistical work, while remaining manageable for exploratory and comparative analysis.

The use of daily data is especially important in this context because the clients decision cannot be supported only by annual means. A retreat location may appear favourable on average and still be problematic in practice if it contains too many rain days, too many very hot afternoons in summer, or too many freezing nights in winter. Daily observations make it possible to build practical indicators that are closer to the business problem.

Each dataset contains 2,981 daily observations for each location, covering the same period and the same set of variables. This structural comparability is essential because it allows the two alternatives to be evaluated under the same analytical framework.

2.2 Dataset Structure and Variables

The two datasets contain daily variables describing thermal, hydrological, wind, solar, and atmospheric conditions. The variables most relevant to the analysis are:

- `temperature_2m_max`: daily maximum temperature;
- `temperature_2m_min`: daily minimum temperature;
- `temperature_2m_mean`: daily mean temperature;
- `precipitation_sum`: daily precipitation total;
- `wind_speed_10m_max`: daily maximum wind speed;
- `wind_gusts_10m_max`: daily maximum wind gust;
- `shortwave_radiation_sum`: daily shortwave solar radiation;
- `sunshine_duration`: daily sunshine duration;
- `relative_humidity_2m_mean`: average daily relative humidity;
- `cloudcover_mean`: average daily cloud cover;
- `sunrise_time`: sunrise time;
- `sunset_time`: sunset time;
- `date`: calendar date;
- `zone`: location identifier.

Although all variables are part of the dataset, not all play the same role in relation to the business problem. The most relevant variables are those that can be clearly connected to guest comfort, operational burden, environmental pleasantness, and climatic stability.

The three temperature variables are central because they capture different aspects of thermal experience. Daily maximum temperature is particularly important for assessing hot summer days and for identifying potentially uncomfortable outdoor conditions during the most commercially attractive season. Daily minimum temperature helps identify freezing nights and operational stress during winter. Daily mean temperature provides a broader sense of seasonal thermal character and supports the interpretation of heating or cooling needs.

Precipitation is one of the most important variables for this project because outdoor retreat activities depend on dry conditions. Total rainfall affects not only whether activities can take place, but also maintenance needs, humidity exposure, and the possible emergence of a more severe low season. A retreat may tolerate moderate cold more easily than repeated wet days if the business model depends strongly on outdoor circulation and atmosphere.

Wind variables matter because stronger winds or gusts reduce thermal comfort, interfere with outdoor programmes, and can create safety concerns in mountain environments. Solar and atmospheric variables, such as sunshine duration, solar radiation, cloud cover, and humidity, matter because they shape the general quality of the environment. A premium wellness retreat benefits from a climate that feels open, bright, and visually pleasant, not only one that is technically acceptable in thermal terms.

To translate these variables into a structured business interpretation, the analysis groups them into four broader dimensions.

Comfort Dimension

Comfort is captured mainly by temperature and rainfall. Guests are more likely to enjoy outdoor programmes when temperatures are moderate, rainfall is absent or very low, and wind is not excessive. The analysis therefore uses temperature thresholds and rainfall-based indicators to approximate the number of days on which retreat activities could comfortably take place.

A particularly useful working indicator is the number of **OK days**, defined as days satisfying:

$$\begin{aligned} \text{precipitation_sum} &< 1.0 \text{ mm,} \\ \text{wind_gusts_10m_max} &< 40, \\ 18^{\circ}\text{C} &\leq \text{temperature_2m_max} \leq 30^{\circ}\text{C.} \end{aligned}$$

This indicator does not claim to measure all aspects of comfort perfectly, but it is analytically useful because it approximates the number of days in which most outdoor retreat activities could take place under acceptable conditions.

Operations and Risk Dimension

Operations and risk are captured mainly by extreme cold, extreme heat, and rainfall burden. Frequent freezing nights increase heating needs and make winter operations more demanding. Very high summer temperatures and warm nights may reduce comfort

when nature-based demand is expected to be strongest and may require stronger cooling systems. Persistent rainfall increases maintenance costs, waterproofing requirements, humidity-related deterioration, and the risk of a more constrained low season.

Atmospheric Quality and Perceived Wellness Environment

The atmosphere dimension is captured mainly by sunshine duration and cloud cover, supported by solar radiation and humidity. This dimension is important because the client is not simply looking for a place where activities are technically feasible. The retreat experience also depends on the visual, emotional, and restorative quality of the environment.

A location with clearer skies and slightly more sunshine may be more aligned with the image of tranquillity, openness, scenic beauty, and psychological well-being expected from a premium wellness retreat. For this reason, the report treats sunshine and cloud cover not as secondary decorative variables, but as meaningful components of the decision.

Predictability Dimension

Predictability concerns the stability of climatic patterns across months and years. This is essential because the client needs more than acceptable average conditions. It needs a location whose summers are reliably similar, whose wet and dry seasons are recognisable, and whose environmental conditions do not fluctuate so sharply that long-term planning becomes difficult.

Summer should be analysed separately, especially June, July, and August, to verify whether warm-season conditions repeat consistently from year to year. Rainfall predictability should also be analysed in two complementary ways: seasonal predictability and day-to-day predictability.

Seasonal predictability asks whether wet and dry months repeat consistently every year. Day-to-day predictability asks whether rain tends to occur in spells or clusters, which matters directly for planning outdoor activities. Later, heatmaps, boxplots, transition probabilities, and dry- or wet-streak distributions can be used to deepen the analysis.

2.3 Data Quality and Initial Limitations

The initial inspection suggests that both datasets are structurally complete and suitable for direct comparison. The relevant variables appear in both locations with the same

format, the time horizon is aligned, and the daily level of detail allows the data to be aggregated by month, season, and year. In practical terms, this means the data are immediately usable for descriptive comparison.

An important strength of the dataset is that it allows the analysis to move beyond broad averages and toward practical indicators that are closer to the clients business needs. Instead of asking only which place is warmer or cooler on average, the report can begin to quantify issues such as the number of acceptable outdoor days, the number of freezing days, the frequency of warm nights, the burden of rainfall, and the atmospheric conditions associated with scenic quality.

The working analysis highlights several useful indicators that emerge naturally from the daily data:

- number of “OK days” for outdoor retreat activities;
- number of days above 30°C;
- number of warm nights above 20°C;
- number of freezing days below 0°C;
- number of rain days;
- atmospheric quality through sunshine duration and cloud cover;
- future predictability measures based on year-to-year variation.

Table 2.1 summarises the main indicators that emerged from the working analysis.

Table 2.1: Comparison of key climatic indicators

Indicator	Guarda	Vila Real
OK days	623	612
Days above 30°C	159	270
Warm nights above 20°C	15	75
Freezing days below 0°C	326	67
Rain days	856	1039
Average annual precipitation (mm)	809	1372

A relevant methodological limitation appears early in the analysis. Since Vila Real is considerably rainier than Guarda in absolute terms, predictability cannot be judged only through total rainfall. A city with more rainfall will naturally have higher totals, but that

does not automatically mean it is less predictable. For this reason, relative measures of variation and month-by-month consistency are needed in order to compare fairly whether each place is operationally stable.

This limitation is important because it prevents the analysis from confusing “more rain” with “more unpredictability”. The client needs both dimensions to be assessed separately. A location may be wetter but seasonally regular, or drier but more erratic. The report therefore recognises from the outset that it must compare not only rainfall volume, but also the consistency of rainfall behaviour across years and the persistence of dry or wet spells.

This report therefore uses the data not only to describe each location, but also to support a concrete investment decision. The analytical emphasis is on evidence that changes the client’s decision: seasonal OK-day timing, rainfall burden, heat and cold risks, atmospheric quality, and the statistical reliability of the observed differences.

Exploratory Orientation

Initial inspection showed that the decision should not be framed as a simple warm-versus-cold comparison. The relevant trade-off is more specific: Guarda is colder in winter but generally drier, brighter, and less exposed to peak-season heat, while Vila Real is milder in winter and spring but wetter and hotter in summer. The detailed figures are therefore presented in the Results chapter, where each visual is interpreted directly in relation to the business decision.

Chapter 3

Methodology

3.1 Preprocessing and Data Preparation

Data Loading and Integration

The analysis uses two raw CSV files obtained from the Open-Meteo Archive API, one for Guarda and one for Vila Real, each containing 2,981 daily observations covering the period from 1 January 2018 to early 2026. Both files share the same structure and variables, which makes them directly comparable without additional format conversion.

The two files were loaded separately and then joined into a single analytical dataset. The `zone` column, already present in the raw data and taking the values `GUARDA` and `VILA_REAL`, was preserved as the city identifier throughout the analysis. After integration, the combined dataset contained 5,962 rows.

Missing Value Assessment

Before any transformation, both datasets were inspected for missing values. The inspection confirmed that neither dataset contains missing values across the 17 variables considered in the project. No imputation or row removal was therefore required. This is an important strength of the dataset, since all subsequent counts, averages, and comparisons are based on the full observation period.

Variable Creation: Temporal Dimensions

Three temporal variables were created from the original time field:

- **month**: the calendar month, used for monthly aggregation and seasonal profile analysis;
- **year**: the calendar year, used for inter-annual consistency analysis;
- **season**: a categorical variable assigning each observation to one of four meteorological seasons — Winter (December–February), Spring (March–May), Summer (June–August), and Autumn (September–November).

These variables allow the analysis to be organised by month, season, and year, which is essential for examining climatic patterns at different temporal scales.

Variable Creation: Unit Conversion

The raw **sunshine_duration** variable is recorded in seconds per day. To facilitate interpretation, it was converted into hours per day:

$$\text{sunshine_hours} = \frac{\text{sunshine_duration}}{3600}$$

All subsequent references to sunshine in the analysis use this derived variable expressed in hours.

Variable Creation: Analytical Indicators

To operationalise the business dimensions identified in the problem framing, five binary indicators were created. Each indicator takes the value 1 when the condition is met and 0 otherwise, allowing direct aggregation into counts of days.

OK Day (ok_day**)**: a day is classified as suitable for outdoor retreat activities if all of the following conditions are simultaneously satisfied:

$$\begin{aligned} \text{precipitation_sum} &< 1.0 \text{ mm,} \\ \text{wind_gusts_10m_max} &< 40, \\ 18^{\circ}\text{C} &\leq \text{temperature_2m_max} \leq 30^{\circ}\text{C.} \end{aligned}$$

This composite indicator approximates days on which outdoor activities can take place under acceptable thermal and meteorological conditions.

Hot Day (hot_day**)**: days on which the daily maximum temperature exceeds 30°C. This threshold captures conditions likely to reduce comfort and increase cooling needs.

Freezing Day (`freezing_day`): days on which the daily minimum temperature falls below 0°C. These conditions are associated with winter operational stress, overnight discomfort, and higher heating demand.

Warm Night (`warm_night`): days on which the daily minimum temperature exceeds 20°C. These conditions are associated with uncomfortable sleeping conditions and greater cooling energy demand.

Rain Day (`rain_day`): days on which daily total precipitation is at least 1.0 mm. This threshold captures meaningful rainfall capable of affecting outdoor programming.

Aggregations

Monthly, seasonal, and annual aggregations were produced by grouping combinations of `month`, `season`, and `year` together with `zone`. For binary indicators, the aggregation function used was *Sum*, which directly yields day counts. For continuous meteorological variables, the main aggregation functions used were *Mean* and, where relevant, *Sum*.

These prepared variables and aggregated tables form the basis for all analyses presented in Chapter 4.

3.2 Exploratory Analysis Strategy

The methodological strategy was organised around four complementary analytical dimensions: temperature, atmosphere, precipitation, and predictability. Together, these dimensions reflect the main business concerns identified in the problem framing, namely guest comfort, operational risk, environmental quality, and long-term planning reliability.

Temperature Dimension

The temperature dimension was analysed to assess the thermal profile of each location throughout the year and to identify exposure to relevant extremes.

The main variables used were `temperature_2m_mean`, `temperature_2m_max`, `temperature_2m_min`, together with the derived indicators `hot_day` and `freezing_day`. Monthly and seasonal aggregations were used to examine how temperatures evolve across the year, while the extreme indicators were used to quantify the frequency of potentially problematic thermal conditions.

The temperature analysis was designed to answer three main questions. First, whether

one location is structurally warmer or colder than the other across the annual cycle. Second, whether the differences are concentrated in specific seasons or persist throughout the year. Third, whether the warmer location is also more exposed to heat stress during the commercially important summer months, and whether the colder location is more exposed to winter operational penalties.

In addition to averages, temperature distributions were analysed using boxplots in order to assess dispersion, overall range, and the presence of extreme observations. This is important because average temperatures alone do not fully describe the likelihood of unusually cold or hot days, which may be operationally relevant even if seasonal means appear moderate.

Overall, the temperature dimension provides the thermal backbone of the comparison, helping distinguish between winter cold exposure, summer heat exposure, and general year-round thermal character.

Atmosphere Dimension

The atmosphere dimension covers the variables most directly associated with the visual and psychological quality of the environment: sunshine duration, cloud cover, and short-wave radiation.

Sunshine duration was analysed using the derived variable `sunshine_hours`. This variable captures the number of hours per day during which direct solar irradiance reached the measurement point, making it a meaningful indicator of perceived brightness and openness. The analysis included overall daily mean and median sunshine duration for each location, monthly breakdowns to identify seasonal patterns, and annual totals to assess year-to-year consistency.

Two threshold-based indicators were also considered: the number of days with more than eight hours of sunshine, interpreted as highly favourable outdoor-light conditions, and the number of days with fewer than two hours of sunshine, interpreted as atmospherically weak or dark days.

Cloud cover was analysed as the complementary variable to sunshine duration. Mean daily cloud cover is expressed as a percentage, where 0% represents a completely clear sky and 100% a fully overcast sky. The analysis followed the same structure as for sunshine: overall summary measures, monthly profiles, and distributional comparison.

To improve interpretability, daily cloud cover values were also grouped into three sky conditions: clear (below 30%), partly cloudy (between 30% and 70%), and overcast (above 70%). These categories allow the comparison to move beyond raw percentages and to

express atmospheric quality in terms that are directly relevant to guest experience.

Shortwave radiation was used as a supporting variable. While sunshine duration captures the duration of direct light, shortwave radiation measures the total solar energy received. Comparing both variables helps confirm whether the directional conclusions about atmospheric quality are internally consistent.

The atmosphere dimension was therefore designed not only to measure brightness, but also to assess whether one location offers a more consistently open, clear, and visually pleasant environmental setting for a premium wellness retreat.

Precipitation Dimension

The precipitation dimension was analysed to compare rainfall burden, rainfall frequency, and temporal variability between the two locations.

The main variables used were **precipitation_sum**, **rain_day**, **month**, **season**, and **year**. The analysis began with overall comparison of daily mean precipitation, total accumulated precipitation, and total number of rain days for each city. These measures establish the absolute rainfall burden associated with each location.

Monthly and seasonal aggregations were then used to examine whether the two cities follow the same general annual rainfall pattern and to identify in which parts of the year the differences are most operationally relevant. This is particularly important because outdoor retreat programming depends not only on how much it rains overall, but also on when the wettest periods occur.

Rainfall frequency was analysed using the binary indicator **rain_day**, since the number of days with meaningful rainfall is often more relevant to retreat operations than annual rainfall totals alone. A location may receive large total rainfall concentrated in a small number of events, or lower totals spread across many wet days; these two situations have different implications for activity disruption and maintenance.

Finally, variability was examined using monthly boxplots, which allow the distribution of rainfall totals to be compared across years within each calendar month. This makes it possible to distinguish between structurally wet months and months whose rainfall is more erratic from year to year.

The precipitation dimension therefore combines level, frequency, and variability in order to assess how strongly rainfall constrains outdoor usability and operational reliability.

3.3 Statistical Summary and Predictability Analysis

The statistical summary and predictability analysis addresses one of the central business-oriented questions defined in the project framework: which location offers more consistent and operationally reliable climatic conditions over time? This question cannot be answered using averages alone. A location may appear acceptable in terms of mean values and still present substantial year-to-year instability that complicates long-term planning.

Descriptive Statistics and Coefficient of Variation

For each key meteorological variable, three summary statistics were computed across the full observation period (January 2018 to early 2026): the mean, the standard deviation, and the coefficient of variation (CV). The CV — defined as the standard deviation divided by the mean and expressed as a percentage — is particularly useful because it allows variability to be compared across variables with different scales and units. A higher CV indicates greater dispersion relative to the mean and therefore lower predictability.

The variables analysed were: daily maximum temperature, daily minimum temperature, daily mean temperature, total precipitation, maximum wind gusts, sunshine hours, and mean cloud cover. Together, these variables cover the four analytical dimensions of the project: comfort, operations and risk, atmosphere, and predictability.

Extreme Event Frequencies

Beyond central tendency and dispersion, the analysis quantified the frequency of extreme events as a percentage of total observation days. Four categories of extreme days were used, consistent with the indicators defined earlier:

- **Freezing days:** daily minimum temperature below 0°C;
- **Hot days:** daily maximum temperature above 30°C;
- **Warm nights:** daily minimum temperature above 20°C;
- **Rain days:** daily precipitation at or above 1.0 mm.

These event frequencies translate raw meteorological conditions into operationally relevant risk indicators.

Annual and Seasonal Consistency

Predictability was assessed at two temporal scales. At the annual level, key indicators — including the number of OK days, freezing days, hot days, rain days, and total precipitation — were computed for each calendar year, excluding 2026 because it is only partially observed. Their year-to-year standard deviations and CVs were then calculated. A location with lower inter-annual CV in operationally relevant indicators can be considered more predictable and easier to plan for.

At the seasonal level, the dataset was divided into the four meteorological seasons and seasonal averages were computed for temperature, precipitation, and OK days. Summer received particular attention because it is the main commercial window for a mountain wellness retreat. Inter-annual variability within summer was assessed by computing the mean maximum temperature and total precipitation for June, July, and August in each year and then calculating the CV across years.

Monthly Inter-Annual Coefficient of Variation

As a complementary measure of predictability, the CV of monthly mean temperature was computed for each calendar month across the observed years. This produces a month-by-month stability profile that reveals not only which season is most variable, but also which specific months present greater planning uncertainty. Such information is useful for identifying whether the most unstable periods coincide with the retreat’s most relevant operational months.

Overall, the predictability analysis complements the thematic climate comparisons by showing not only how Guarda and Vila Real differ on average, but also how stable those differences are across time.

3.4 Statistical Validation Strategy

The validation strategy was designed to separate three related questions. First, the analysis tests whether observed differences between the two locations are statistically distinguishable from random variation. Second, it evaluates whether statistically significant differences are large enough to matter for the investment decision. Third, it checks whether the main direction of the results remains stable after accounting for seasonality.

This distinction is important because the dataset contains 2,981 daily observations per location. With a large number of observations, even small differences can become sta-

tistically significant. For this reason, the report interprets p-values together with effect sizes and business relevance. Welch’s t-tests were used for mean comparisons where appropriate, while Cohen’s d was used to assess the practical size of differences. Seasonal and monthly breakdowns were then used as a control against misleading annual averages.

The validation logic therefore follows a conservative rule: a result is treated as decision-relevant only when it is directionally consistent, operationally meaningful, and not merely statistically significant. Rainfall, summer OK-day concentration, heat exposure, and annual OK-day stability receive the greatest weight because they directly affect outdoor programming, maintenance burden, guest comfort, and revenue reliability.

3.5 Predictive Modelling of OK Days

The predictive model is used as a validation and planning tool, not as the main basis for the location recommendation. Its purpose is to test whether OK-day availability follows a recognisable structure using location, seasonality, and atmospheric variables. If OK days can be predicted from these indirect inputs, the indicator is more useful for scheduling than a purely descriptive count.

The target variable is `ok_day_label`, with two classes: *OK* and *Not OK*. The model uses a planning-oriented feature set:

- `zone`, identifying Guarda or Vila Real;
- `season`, `month`, and `day_of_year`, capturing calendar timing;
- `sunshine_hours`, `cloudcover_mean`, `relative_humidity_2m_mean`, and `shortwave_radiation_sum`, capturing atmospheric and experiential conditions.

A key methodological choice was to exclude `temperature_2m_max`, `precipitation_sum`, and `wind_gusts_10m_max` from the predictive dataset. These variables directly define the OK day rule. Including them would make the model almost circular: the algorithm would simply relearn the definition of the target rather than provide an independent planning-oriented validation.

Four classification approaches were compared: Logistic Regression, Decision Tree, Random Forest, and Naive Bayes. The models were evaluated using stratified cross-validation. Because OK days represent only 1,235 of 5,962 observations, the interpretation prioritises AUC, F1-score, recall, and false positives rather than accuracy alone. The model results are kept secondary in the recommendation because they confirm the structure of the climate patterns rather than changing the location choice by themselves.

Procedural Comparability Check

The model also includes a comparability check between the two locations. In this project, **zone** is not a protected human attribute; it is the business alternative under evaluation. Therefore, the purpose of this check is not to claim social-equity impact, but to verify that the two locations are being evaluated on equal analytical terms.

The comparison is procedurally consistent because both locations use the same period, the same number of observations, the same variables, and the same OK-day definition. If model performance differs by location, that difference should be examined, but it may reflect real climatic structure rather than model bias. In other words, the goal is to compare the locations under the same rules, not to force equal predicted OK-day probability for places that genuinely have different weather patterns.

Chapter 4

Results

4.1 Key Findings

Overall Indicator Counts

Table 4.1 presents the total counts of each binary indicator over the full observation period (2018 to early 2026, 2,981 days per location).

Table 4.1: Summary of key daily indicators: Guarda vs. Vila Real (2018–2026).

Indicator	Guarda	Vila Real	Interpretation
OK Days	623	612	G +11 (slight advantage)
Days above 30°C	159	270	VR nearly double G
Warm nights above 20°C	15	75	VR 5× more
Freezing days below 0°C	326	67	G nearly 5× more
Rain Days (≥ 1 mm)	856	1039	VR +183 more

The total number of OK days is nearly identical between the two locations (623 vs. 612), which means that the aggregate comfort opportunity is comparable at the annual level. However, this surface-level similarity conceals different seasonal structures and risk profiles, which are revealed by the breakdown below.

Monthly Profile of OK Days

Figure 4.1 shows the number of OK days per calendar month for both locations.

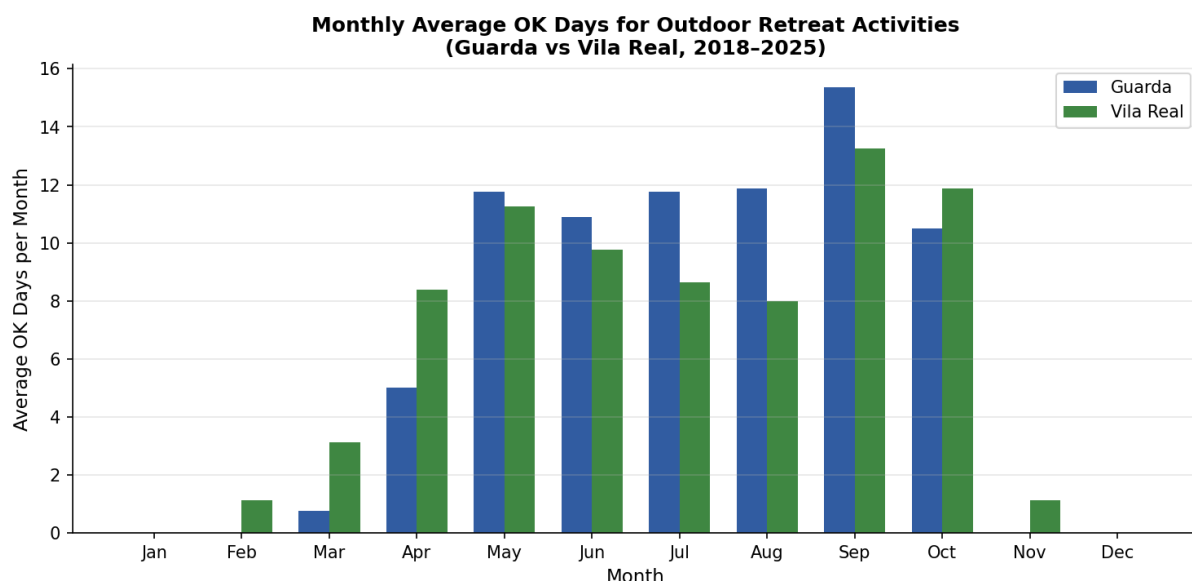


Figure 4.1: Total OK days per calendar month for Guarda and Vila Real (2018–2026).

The monthly breakdown reveals a clear divergence in seasonal structure. Guarda concentrates its OK days strongly in summer: it records 276 OK days during the summer months (June, July, August), which represents 44.3% of its annual total. In contrast, Vila Real accumulates only 211 OK days in the same season (34.5% of its annual total), because many summer days are excluded by the 30°C upper temperature limit.

During spring, the pattern reverses. Vila Real records 182 OK days in spring compared to 140 for Guarda, because spring temperatures in Vila Real more frequently fall within the 18–30°C comfort band while Guarda is still too cold. In winter, both locations record virtually no OK days (Guarda: 0; Vila Real: 9), confirming that winter is a structurally difficult season for both retreats.

The practical implication is that the two locations offer different types of commercial calendar. Guarda provides a more concentrated summer season for outdoor programming, while Vila Real extends usable conditions into spring but sacrifices summer capacity due to heat.

Annual Consistency of OK Days

Figure 4.2 presents the total OK days per year from 2018 to 2025 (2026 is excluded as a partial year).

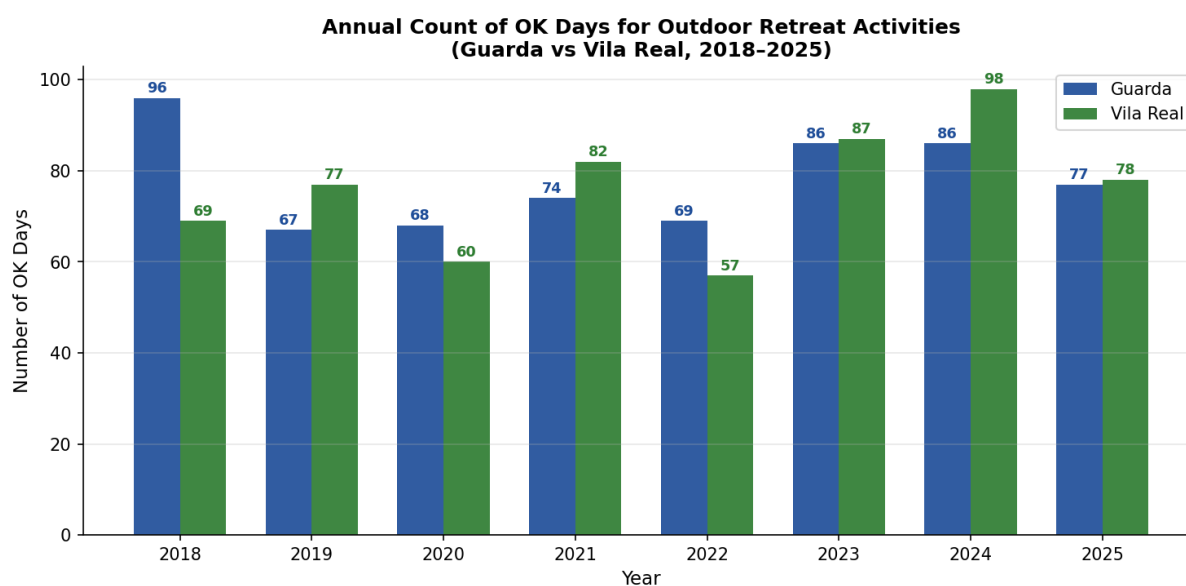


Figure 4.2: Annual total of OK days for Guarda and Vila Real (2018–2025).

Year-to-year variation is substantial in both locations. Guarda ranges from 67 OK days (2019) to 96 (2018), while Vila Real ranges from 57 (2022) to 98 (2024). Neither location shows a stable year-on-year trend, which reflects the inherent variability of meteorological conditions. Guarda leads in five of the eight years (2018, 2020, 2021, 2022, 2025), which confirms that the overall advantage of Guarda is modest and not entirely consistent.

Monthly Profile of Extreme Events

Figure 4.3 presents the monthly distribution of extreme heat days, freezing days, and rain days for both locations.

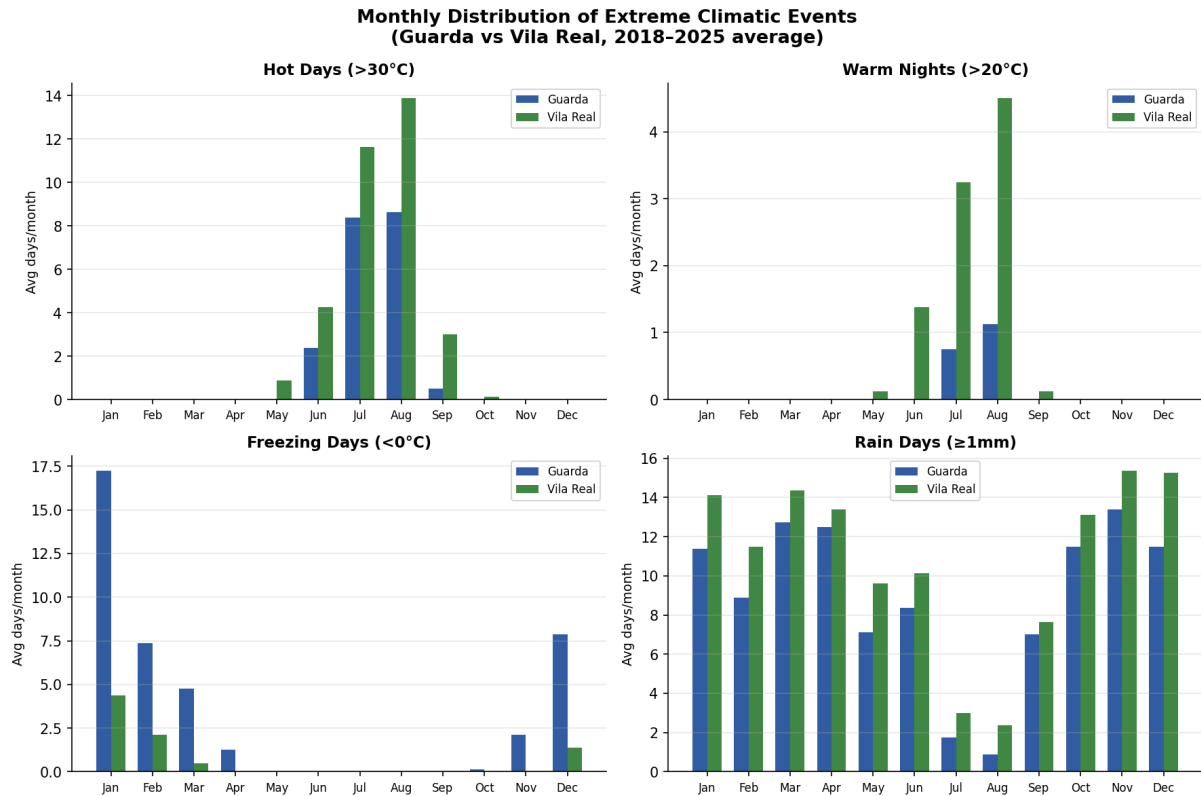


Figure 4.3: Monthly counts of extreme heat days (top left), warm nights (top right), freezing days (bottom left), and rain days (bottom right) for Guarda and Vila Real (2018–2026).

The extreme heat comparison shows that Vila Real exceeds 30°C far more frequently than Guarda. Over the full period, Vila Real records 270 extreme heat days (9.1% of all days) versus 159 in Guarda (5.3%). The difference is concentrated in July and August. The freezing day comparison shows the opposite pattern. Guarda records 326 freezing nights (10.9% of all days) versus only 67 in Vila Real (2.2%), concentrated mainly in January and February. Rain days follow a consistent pattern: Vila Real is wetter in every month of the year, with 183 additional rainy days over the observation period.

Seasonal Breakdown

Figure 4.4 summarises the four key indicators broken down by season for each location.

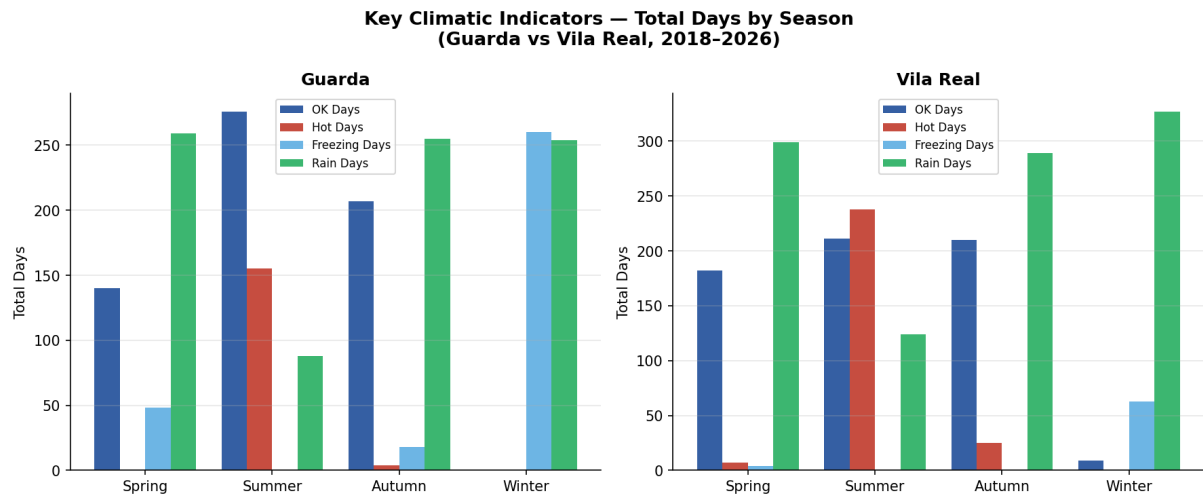


Figure 4.4: Seasonal breakdown of OK days, extreme heat days, freezing days, and rain days for Guarda (left) and Vila Real (right).

The seasonal breakdown clarifies the distinct operational calendars of the two locations. In summer, Guarda has a clear advantage in OK days (276 vs. 211) and far fewer extreme heat events (155 vs. 238), making it a more reliable and comfortable platform for the core outdoor programming season. In autumn, the two locations are almost equally matched on OK days (207 vs. 210), though Vila Real remains substantially rainier.

In winter, Guarda's 260 freezing days create a structurally harder operational environment compared to Vila Real's 63, confirming that winter operations in Guarda would require significantly greater investment in heating infrastructure and guest experience management.

Key Findings - Temperature

The temperature analysis adds precision to the comfort results by isolating the thermal component from rainfall and wind. Vila Real is consistently warmer across all months and seasons, but this does not automatically imply greater suitability.

Monthly Temperature Analysis

Temperature is a fundamental factor in determining the overall comfort and usability of a location throughout the year.

The monthly analysis shows a clear seasonal pattern in both locations. Temperatures increase progressively from winter to summer, peaking in July and August, and decrease towards winter. Across all months, Vila Real consistently presents higher average temperatures than Guarda.

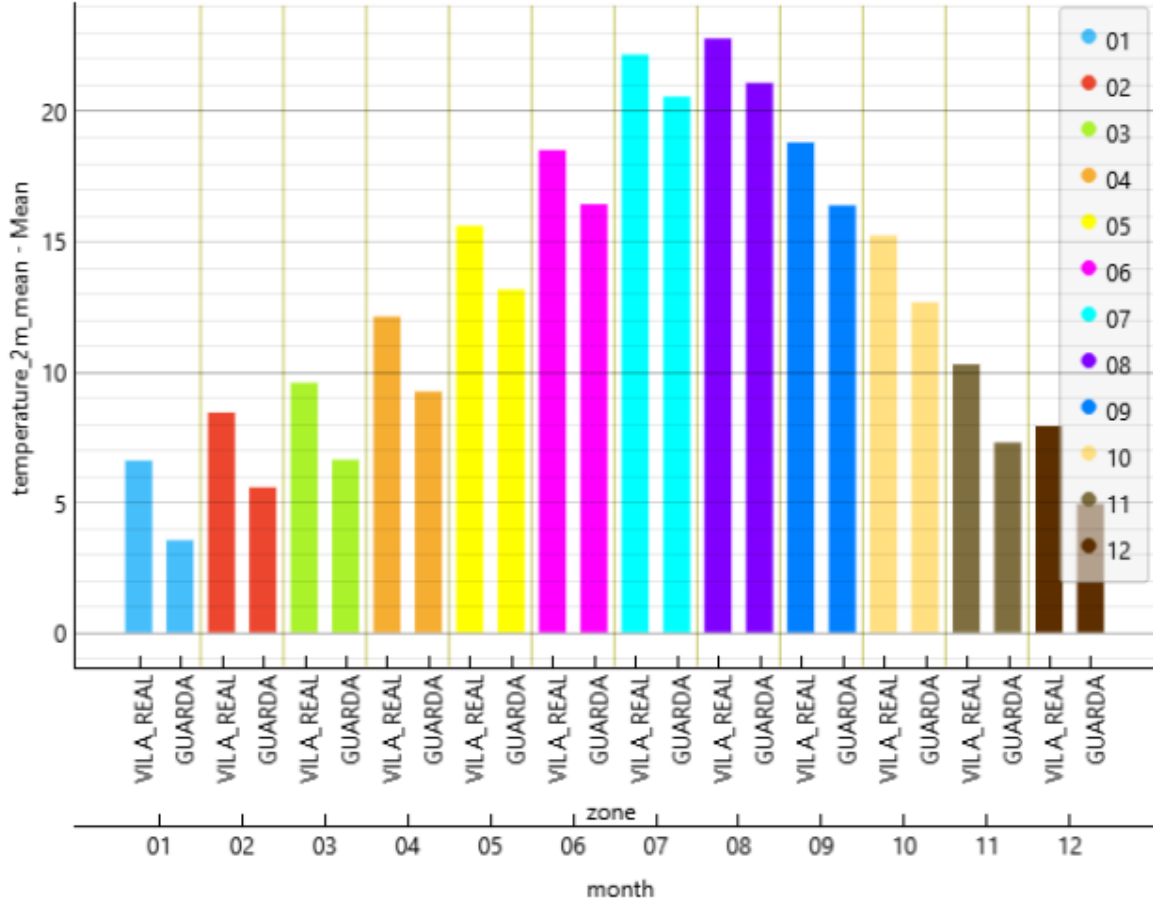


Figure 4.5: Average monthly temperature in Vila Real and Guarda.

Figure 4.5 confirms that both locations follow similar seasonal dynamics, with a systematic temperature gap between them. This suggests that Vila Real provides more favourable thermal conditions throughout the year, particularly during transitional seasons, while Guarda may present limitations due to lower temperatures.

Temperature Distribution and Variability

Beyond average values, it is important to analyse the distribution of temperatures in order to understand variability and the presence of extreme observations.

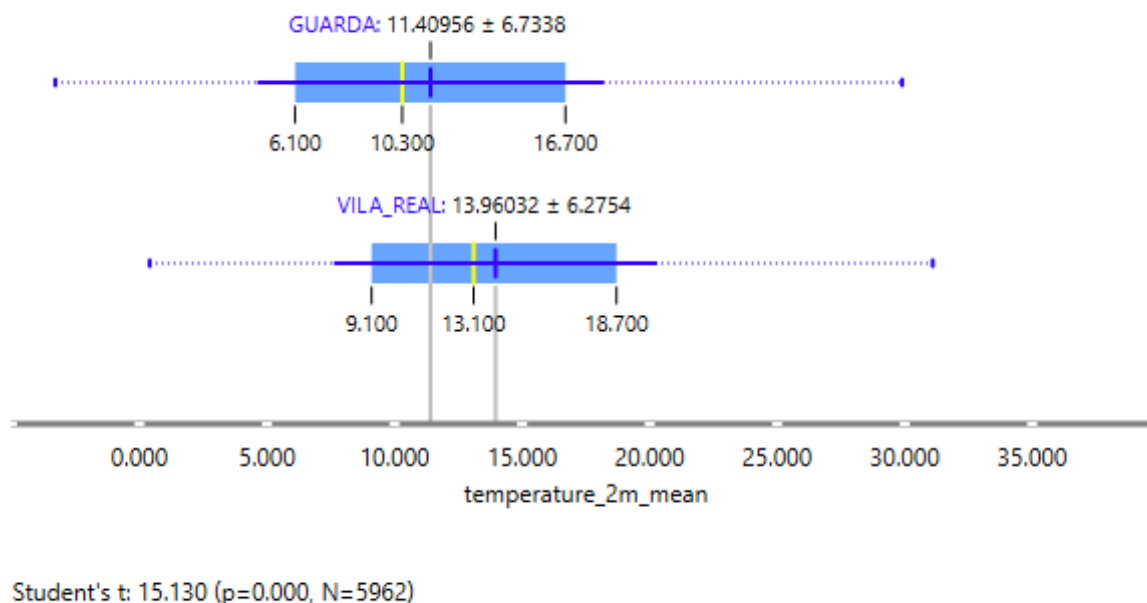


Figure 4.6: Distribution of temperature values for both locations.

Vila Real shows higher overall temperature levels, while Guarda presents lower minimum values. The variability is comparable, although Guarda exhibits more extreme low-temperature observations. This suggests a higher likelihood of cold extremes, reinforcing its less stable thermal profile.

Key Findings Atmosphere (Sunshine and Cloud Cover)

Across the full observation period, Guarda consistently exhibits a brighter atmospheric profile than Vila Real. The mean daily sunshine duration is 9.12 hours in Guarda versus 8.82 hours in Vila Real, while mean daily cloud cover is 51.2% in Guarda and 54.1% in Vila Real.

Monthly Sunshine and Cloud Cover Profile

Figure 4.7 presents the average daily sunshine duration for each calendar month and the corresponding average monthly cloud cover.

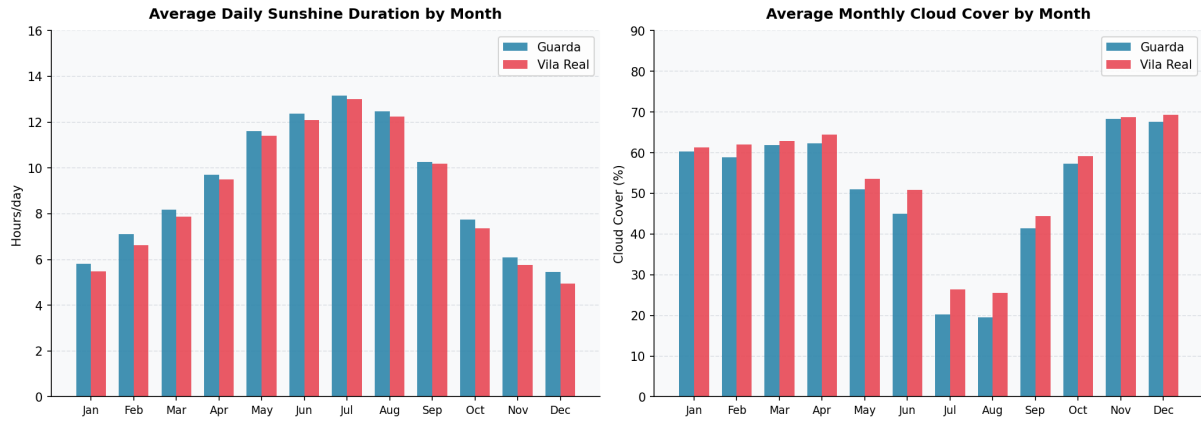


Figure 4.7: Average daily sunshine duration (left) and average monthly cloud cover (right) for Guarda and Vila Real.

Guarda records more sunshine than Vila Real in every month of the year. During the summer peak months of June, July, and August, Guarda averages 12.67 hours of sunshine per day versus 12.46 hours for Vila Real. Summer cloud cover in Guarda is 28.0%, compared to 34.0% in Vila Real.

The monthly cloud cover series confirms that Guarda's advantage is not limited to one season. Guarda has lower average cloud cover in every month of the year, with the largest gaps in July and June.

Distribution of Sky Conditions

Figure 4.8 presents the distribution of daily cloud cover values and the proportion of days in each sky condition category.

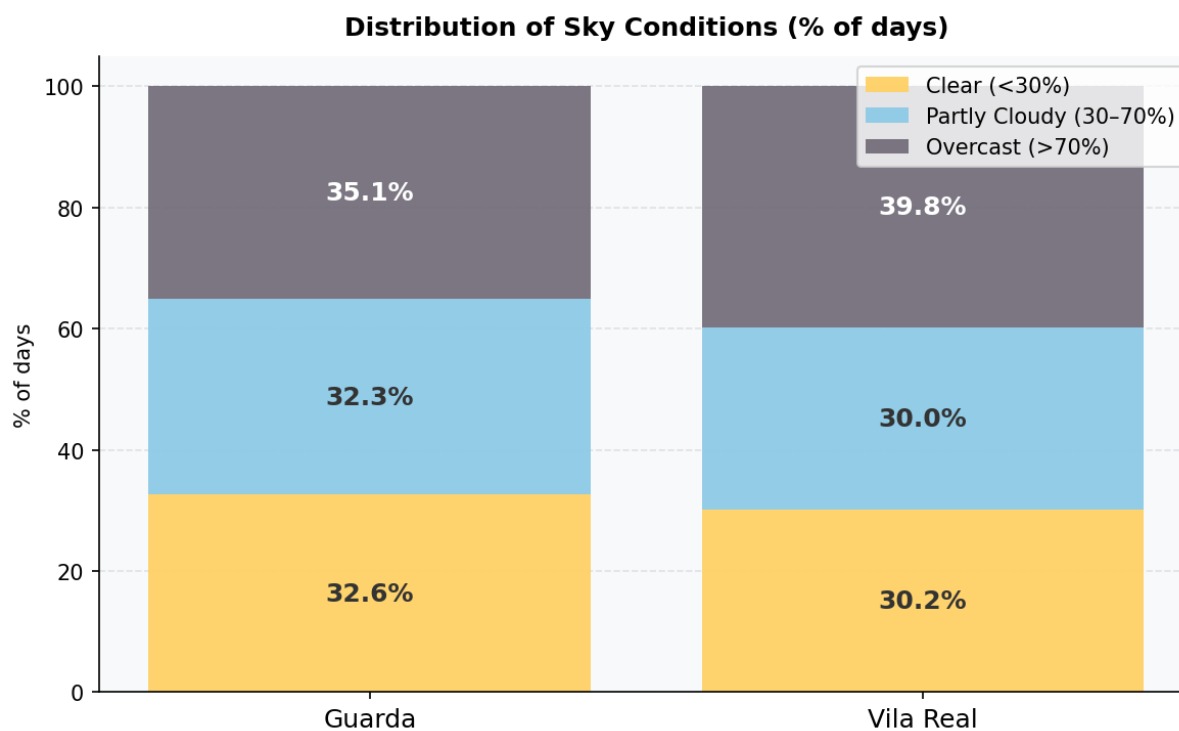


Figure 4.8: Distribution of daily cloud cover values (left) and proportion of days by sky condition category (right).

Guarda records 32.6% of days as clear (cloud cover below 30%), compared to 30.2% for Vila Real. At the other extreme, Guarda has 35.1% of days classified as overcast (above 70%), versus 39.8% for Vila Real. In terms of high-sunshine days (above 8 hours), Guarda records 69.8% of all days, compared to 67.6% for Vila Real. Low-sunshine days (below 2 hours) are less frequent in Guarda (8.0%) than in Vila Real (10.7%).

Summary Table: Atmospheric Indicators

Indicator	Guarda	Vila Real
Mean daily sunshine (h/day)	9.12	8.82
Mean daily cloud cover (%)	51.2%	54.1%
Avg. annual sunshine total (h)	~3,350	~3,250
Clear days / year (<30% cloud)	32.6%	30.2%
Overcast days / year (>70% cloud)	35.1%	39.8%
High-sunshine days (>8 h)	69.8%	67.6%
Low-sunshine days (<2 h)	8.0%	10.7%
Summer cloud cover (Jun–Aug)	28.0%	34.0%
Mean solar radiation (MJ/mš/day)	16.29	15.57

Summary of key atmospheric indicators for Guarda and Vila Real.

Figure 4.9 consolidates the atmospheric evidence into a single visual summary. The chart is useful because it shows that Guarda’s advantage is not dependent on one isolated indicator: it combines more sunshine, lower cloud cover, fewer overcast days, and fewer low-sunshine days. This supports the interpretation that Guarda offers a more consistent open-air wellness atmosphere.

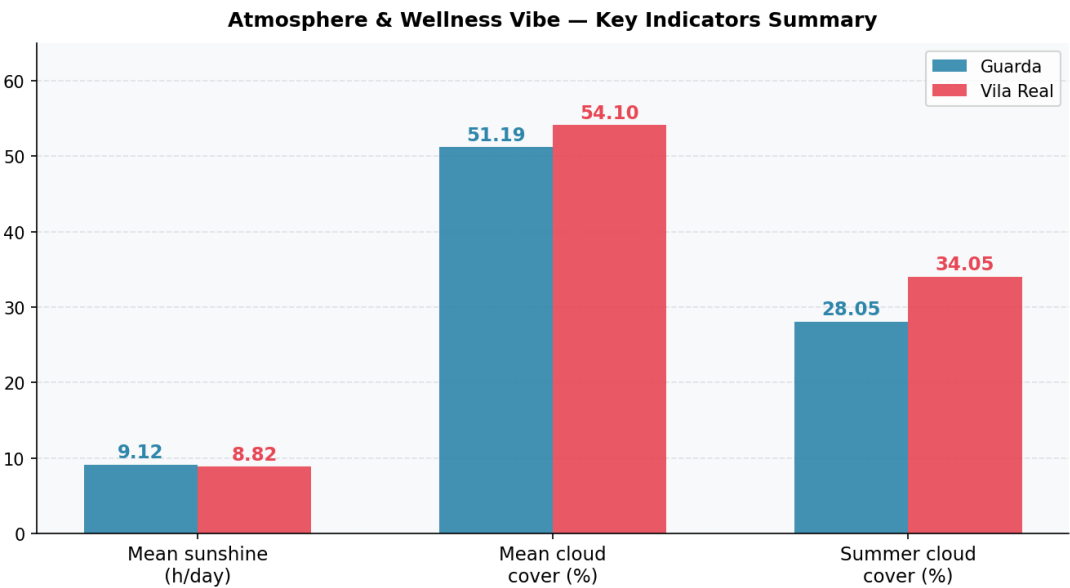


Figure 4.9: Summary of atmospheric indicators relevant to wellness experience.

Key Findings - Precipitation

Overall Precipitation Comparison

The precipitation analysis shows a consistent trade-off between the two locations: Vila Real is clearly wetter, while Guarda is drier but not necessarily more regular in relative terms.

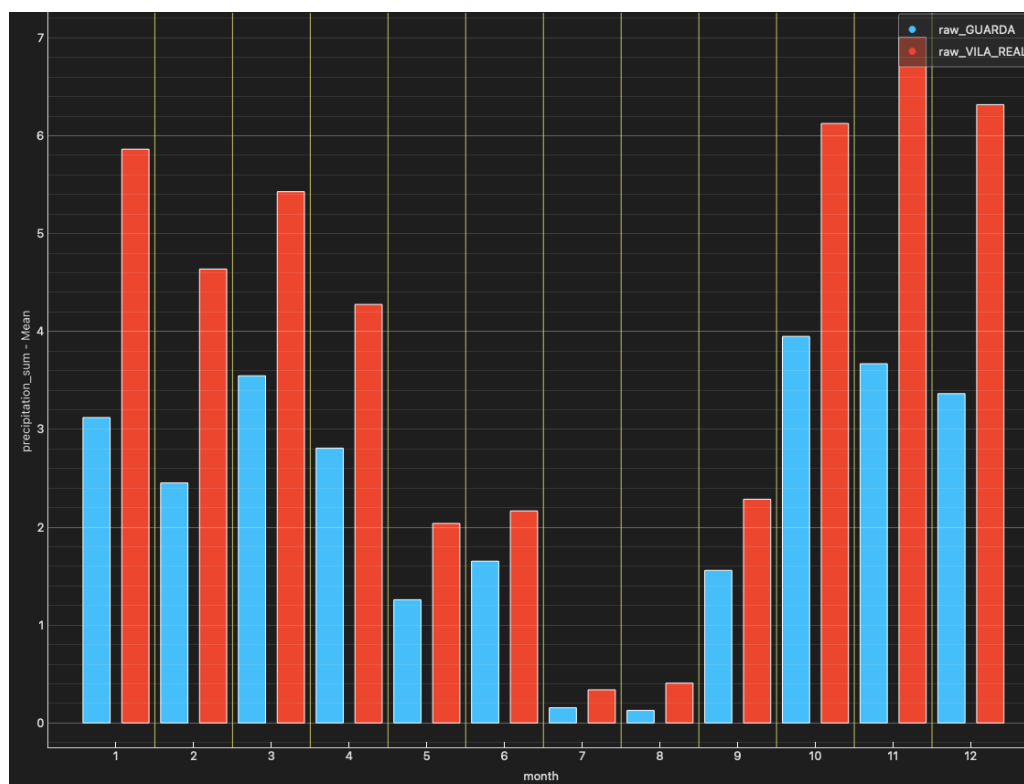


Figure 4.10: Distribution of daily precipitation (mm) for Guarda and Vila Real. Student's t : 8.330 ($p = 0.000$, $N = 5962$).

Vila Real has a higher average daily precipitation, about 3.93 mm/day, compared with 2.31 mm/day in Guarda. It also has a much higher total accumulated precipitation over the full period, 11,714.1 mm versus 6,896.4 mm, and more rain days, 1,039 against 856.

The boxplot reinforces this pattern. Vila Real not only has a higher mean precipitation, but also a higher standard deviation (8.88 vs. 5.78 in Guarda). A useful nuance comes from the coefficient of variation: Guarda has a precipitation CV of about 2.50, while Vila Real has 2.26. This shows that “less rain” and “more predictability” are not exactly the same thing.

Monthly and Seasonal Precipitation Patterns

The monthly and seasonal charts show strong seasonality in both cities. In both locations, summer is the driest period, while autumn and winter are the wettest. However, Vila Real is wetter in almost every month and in all four seasons.

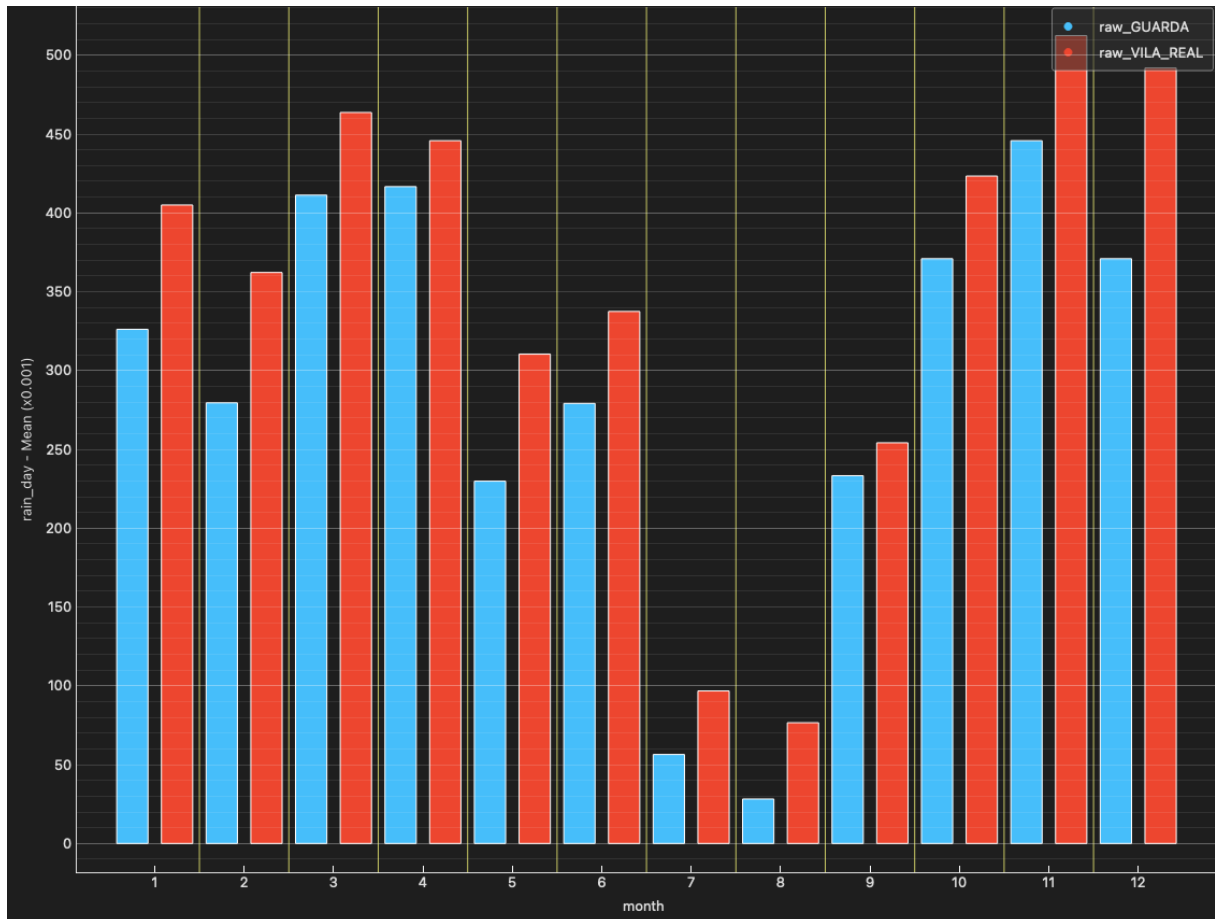


Figure 4.11: Average daily precipitation (mm) by month for Vila Real and Guarda.

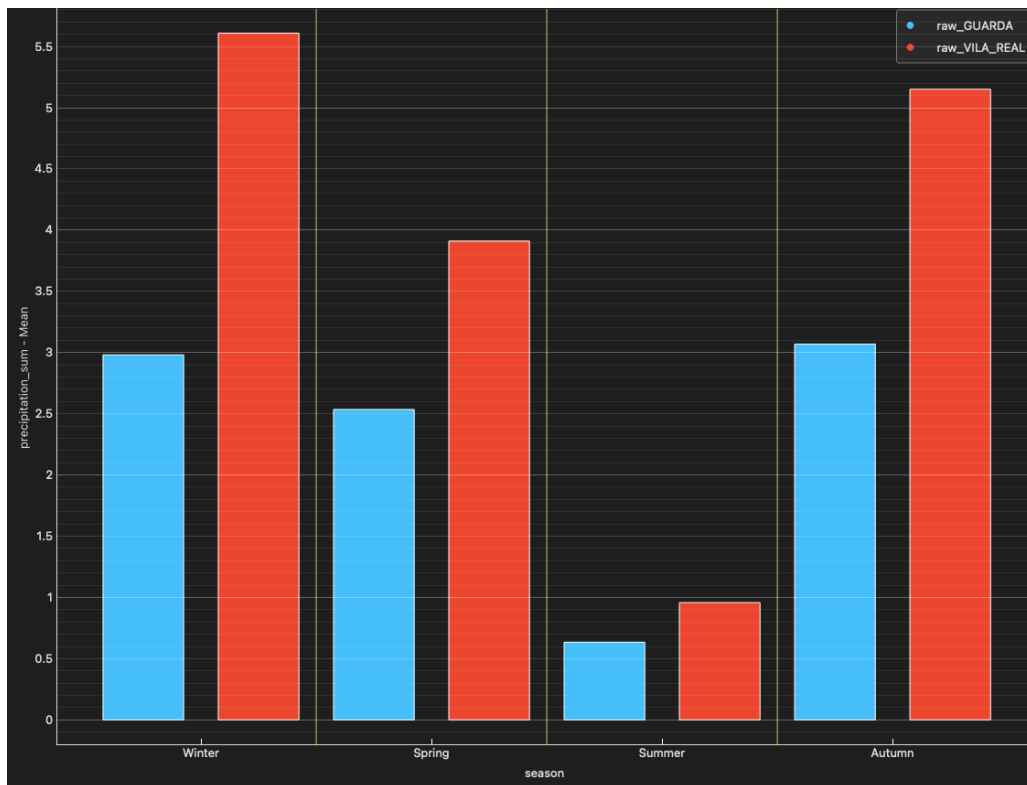


Figure 4.12: Average daily precipitation (mm) by season for Vila Real and Guarda.

Figure 4.11 confirms that the difference is not that one city has rain in different seasons, but that Vila Real intensifies the same wet-season pattern across every month of the year.

Rainfall Frequency

From an operational standpoint, rainfall frequency is more actionable than rainfall total, because guests experience cancellations through rainy days rather than through annual millimetres. Vila Reals higher rain-day counts imply a structurally greater chance of disrupted outdoor activities.

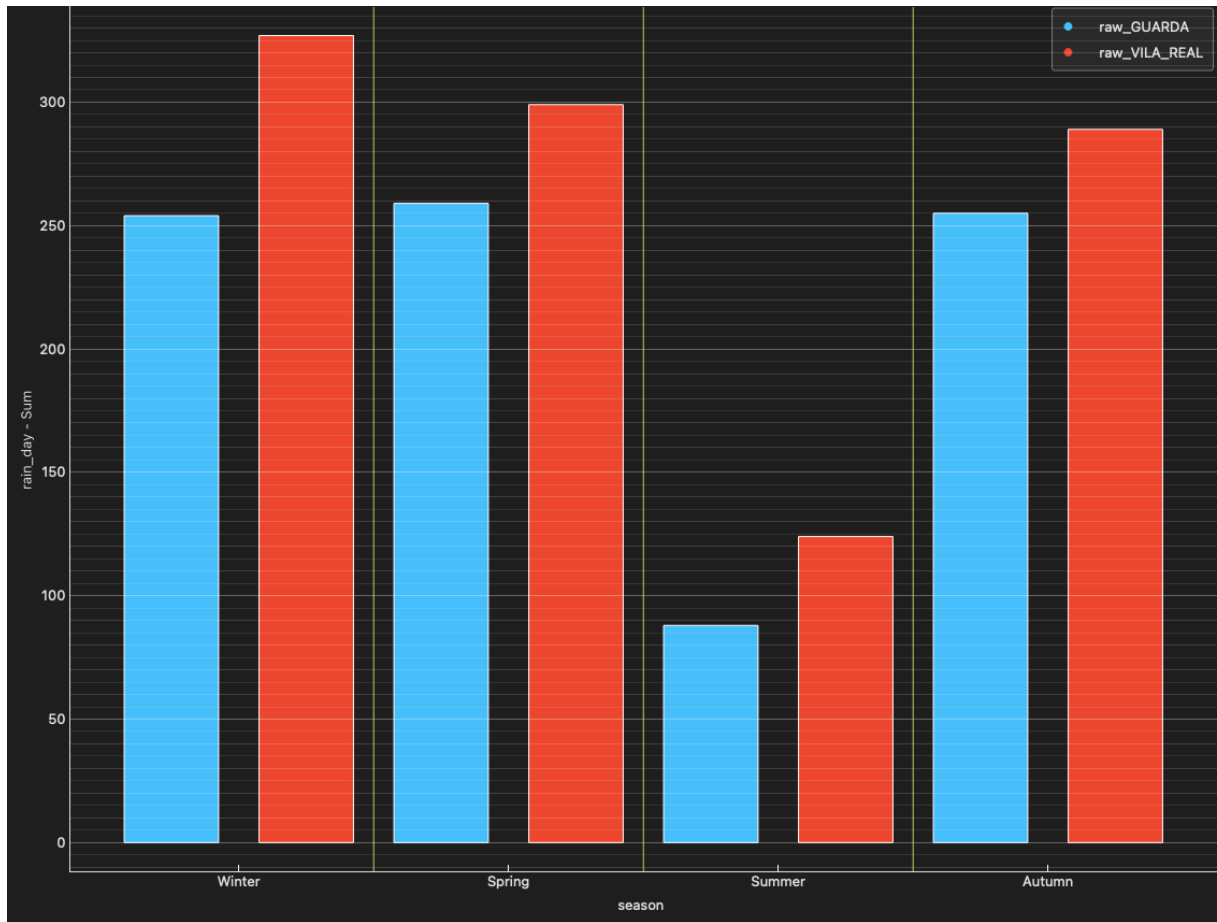


Figure 4.13: Rain-day frequency by season for Vila Real and Guarda (sum across the full observation period).

Figure 4.14 summarises the precipitation validation. It confirms that the rainfall gap is not only a matter of isolated extreme days: Vila Real is wetter in volume, frequency, and seasonal burden. This is one of the most important pieces of evidence in the recommendation because it directly affects outdoor schedule reliability, maintenance exposure, and the perceived quality of the retreat environment.

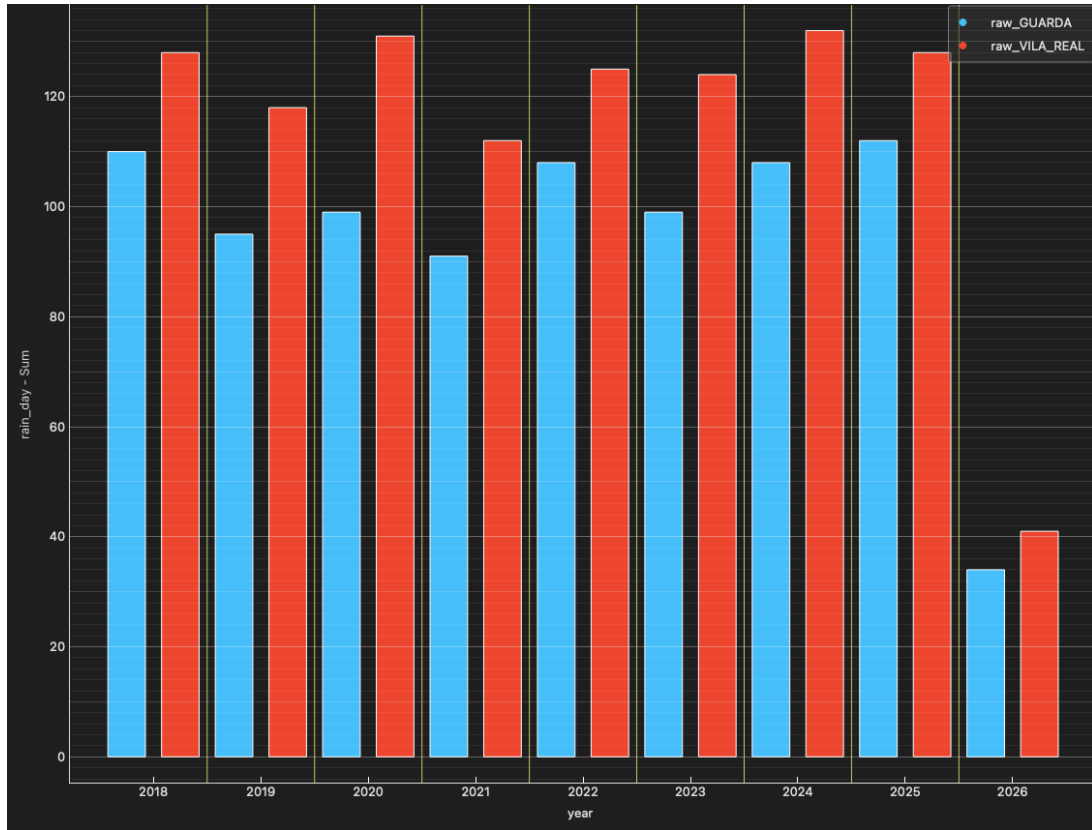


Figure 4.14: Summary of precipitation burden and validation indicators.

Monthly Variability

The monthly boxplot for Guarda shows that precipitation totals are generally low and stable during summer, while the wetter months display wider interquartile ranges and longer whiskers. The monthly boxplot for Vila Real shows substantially larger ranges throughout the year, particularly in the wettest months. This confirms that while the seasonal pattern is predictable in direction, the wet-season amplitude is much greater in Vila Real.

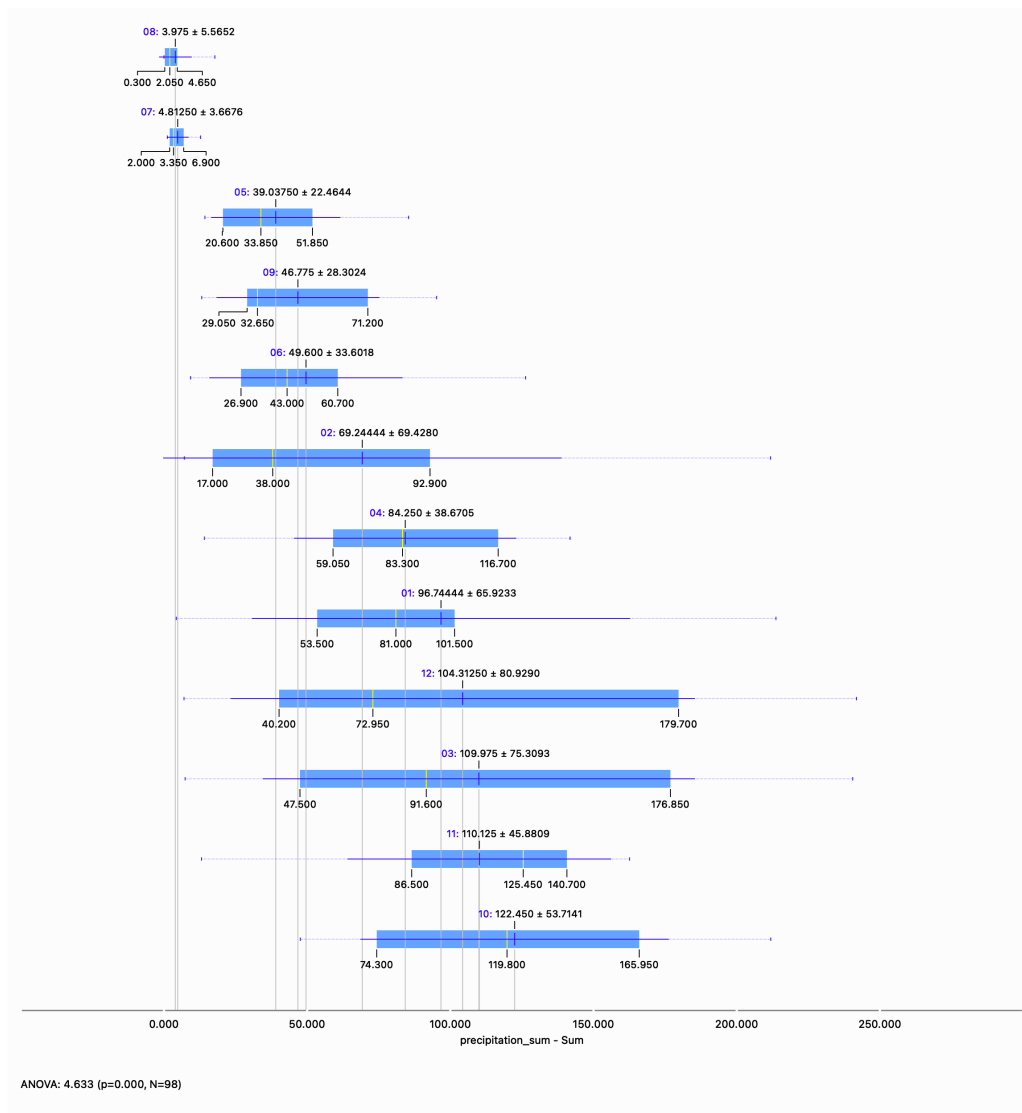


Figure 4.15: Monthly distribution of precipitation totals (mm) for Guarda by month. ANOVA: 4.633 ($p = 0.000$, $N = 98$).

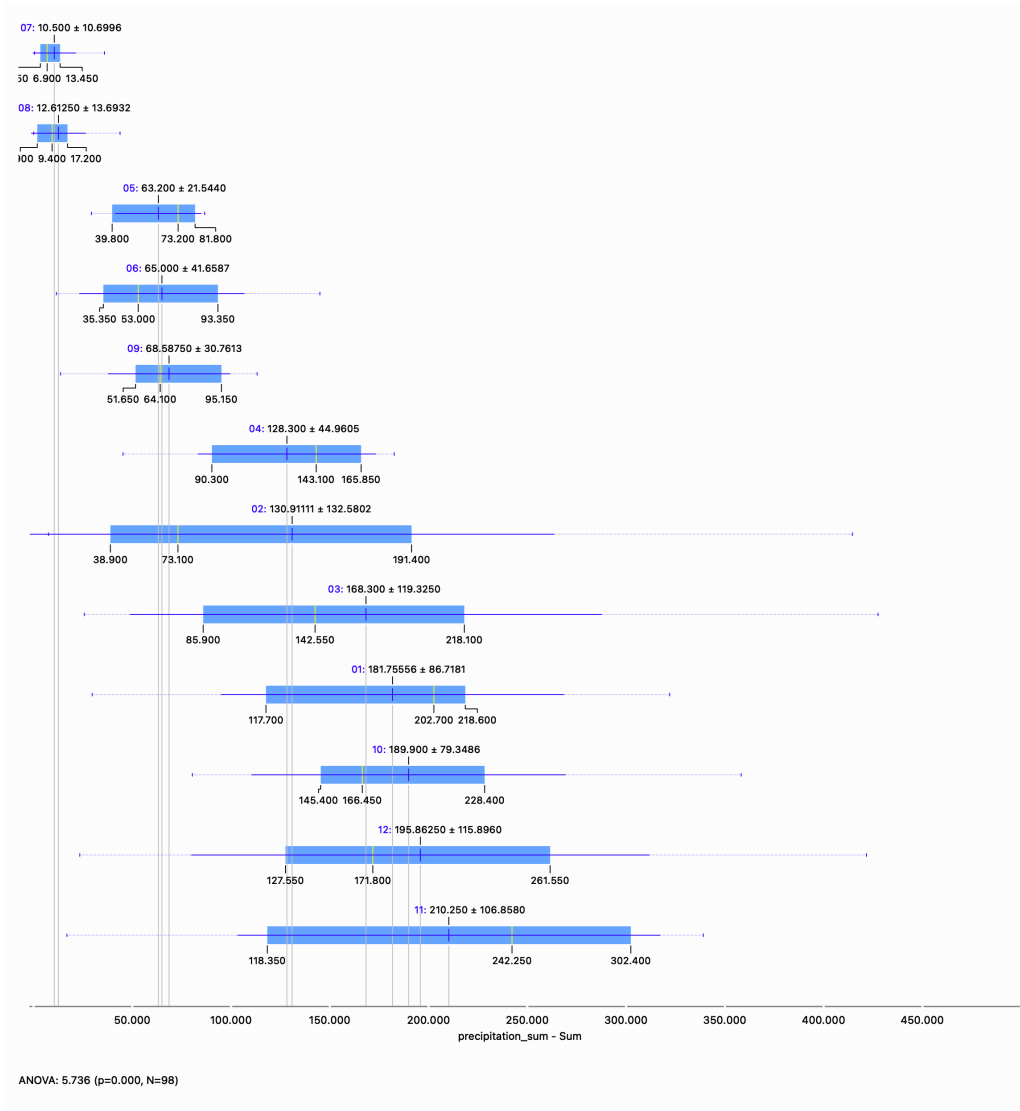


Figure 4.16: Monthly distribution of precipitation totals (mm) for Vila Real by month. ANOVA: 5.736 ($p = 0.000$, $N = 98$).

4.2 Comparison Between Alternatives

Comparison Between Alternatives Comfort Dimension

The indicator analysis establishes the following comparative picture for the comfort dimension.

Aggregate outdoor usability is comparable, but not decision-equivalent. Both locations offer approximately the same number of days per year on which outdoor retreat activities can take place under acceptable thermal and meteorological conditions: 623 in Guarda and 612 in Vila Real. The annual difference is only 11 days, so the analysis should not overstate the aggregate gap. However, the similarity in annual totals is not the same

as a true operational tie, because the business value of an OK day depends strongly on when it occurs in the commercial calendar.

Guarda's comfort advantage is concentrated in summer. Guarda delivers 276 OK days in summer against Vila Real's 211, a difference of 65 suitable summer days. This is more important than the annual difference because summer is likely to be the most relevant period for hiking, yoga, meditation sessions, outdoor dining, and nature-based experiences. In this sense, the comfort dimension is better interpreted as a question of operational calendar than as a simple count of good days.

Vila Real's heat risk undermines its summer usability. The primary reason for Vila Real's lower summer OK day count is the frequency of extreme heat days, which increases cooling demands and restricts outdoor activities during the peak season.

Guarda's winter risk is operational rather than experiential. Guarda's freezing days primarily concern heating costs and winter opening viability rather than peak-season guest comfort.

Vila Real has a structural rainfall disadvantage. With 183 additional rain days over the observation period, Vila Real imposes a greater operational burden on outdoor programming.

In summary, from the comfort perspective, Guarda offers a more favourable platform for a premium summer-oriented outdoor wellness retreat. Its advantage lies not in the total number of comfortable days but in where those days fall in the calendar and in the lower frequency of heat events during the core season. Vila Real's spring advantage remains relevant, especially if the business model later prioritises shoulder-season retreats, but this advantage is less decisive if the final concept depends mainly on peak-season outdoor programming.

Explicit Prioritisation of Decision Factors

To make the decision framework clearer, Table 4.2 ranks the analysed factors by their relevance to a premium mountain wellness retreat. This prioritisation is not a final recommendation; it clarifies which evidence should carry the most weight in the final business decision.

Table 4.2: Priority of decision factors for the Mountain Wellness retreat.

Factor	Priority	Reason for Weighting	Current Advantage
Summer OK days	High	Core period for outdoor wellness demand and nature-based programming.	Guarda
Heat and warm-night risk	High	Directly affects guest comfort, sleep quality, and peak-season scheduling.	Guarda
Rain days and rainfall burden	High	Rain disrupts outdoor activities and increases maintenance risk.	Guarda
Atmospheric quality	Medium–High	Sunshine and lower cloud cover support the premium wellness experience.	Guarda
Winter cold exposure	Medium	Important for operating costs, but mainly affects the lower season.	Vila Real
Spring OK days	Medium	Relevant for shoulder-season programming and extended calendar use.	Vila Real
Rain-day pre-dictability	Medium	Useful for planning, but less important than absolute rain burden.	Vila Real

Comparison Between Alternatives Temperature

The comparison reveals a non-linear trade-off. Vila Reals higher temperatures are advantageous in winter and early spring, when they reduce cold exposure and expand the number of moderately pleasant days. However, the same thermal upward shift becomes less advantageous in summer, where it increases the frequency of days above 30°C and the occurrence of warm nights. Guarda, by contrast, suffers a clearer winter penalty but remains closer to the desired thermal range during peak summer months.

Comparison Between Alternatives Atmospheric Quality and Perceived Wellness Environment

The atmosphere dimension directly shapes the perceived wellness environment of a location. Across every indicator examined, Guarda performs better than Vila Real on the atmospheric dimension. It receives more sunshine on average, has fewer overcast days, more clear days, and accumulates this advantage consistently across the year.

From a guest experience perspective, this translates into better conditions for outdoor meals, views, and open-sky activities. Vila Real remains viable from an atmospheric

perspective, but its profile is less aligned with the aesthetic and psychological expectations attached to a premium mountain retreat.

Comparison Between Alternatives Precipitation

Vila Real is the rainier location by every absolute measure: higher daily mean precipitation, higher total precipitation, more rain days, and higher wet-season values. Guarda is drier and therefore likely offers more favourable conditions for outdoor wellness activities, especially from an operational point of view.

From a precipitation perspective, Guarda appears more suitable for a premium mountain wellness retreat because it is drier overall and has fewer rain days, which favours outdoor activities and reduces disruption risk. Vila Real, while slightly more regular relative to its own wetter baseline, presents a clearly rainier climatic profile.

4.3 Robustness Checks / Validation

The validation results confirm that the main comparison is not driven by isolated visual patterns. The strongest statistically and practically relevant difference is precipitation: Vila Real receives substantially more rain, has more rain days, and carries a larger operational disruption burden. Atmospheric differences are also statistically detectable, but their effect sizes are smaller and should be interpreted as supporting evidence rather than as the main driver of the recommendation.

Table 4.3: Statistical validation of selected location differences.

Variable		Direction		Test	p-value	Decision interpretation
Daily precipitation		Vila higher	Real	$t = 8.329$	< 0.001	Robust and operationally meaningful; effect size $d \approx 0.22$, with Vila Real receiving about 70% more rainfall.
Cloud cover		Vila higher	Real	$t = 3.304$	0.001	Detectable but small practical effect; supports Guarda's clearer atmosphere.
Sunshine duration		Guarda higher		$t = 2.889$	0.004	Detectable but small practical effect; reinforces Guarda's atmosphere advantage.

Figure 4.17 visualises the statistical validation results and helps separate statistical significance from practical importance. The precipitation difference is the most decision-relevant result because it combines statistical strength with clear operational meaning. The sunshine and cloud-cover differences are smaller, but they remain directionally consistent with the atmosphere findings.

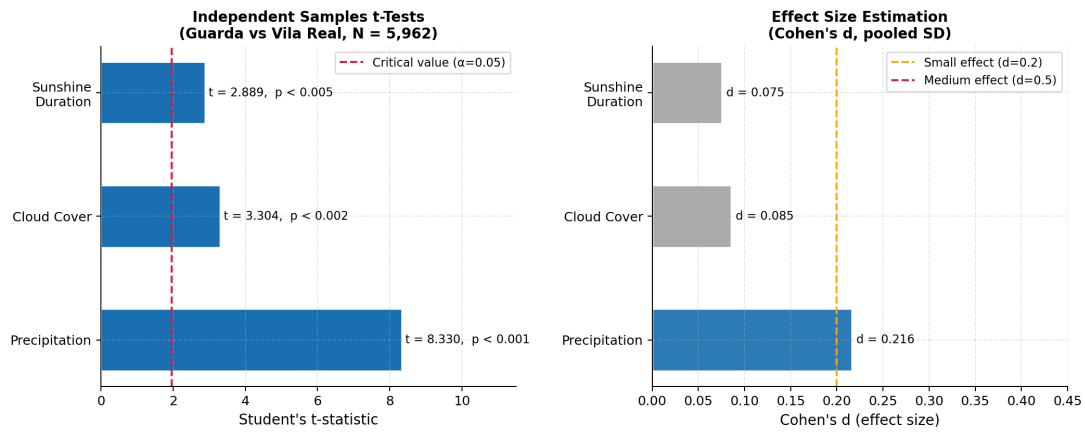


Figure 4.17: Statistical tests and effect-size interpretation for selected location differences.

The precipitation result is the most important validation finding. Vila Real accumulated 11,714.1 mm of rainfall over the observation period compared with 6,896.4 mm in Guarda, or roughly 70% more rainfall. It also recorded 1,039 rain days compared with 856 in Guarda, equivalent to 183 additional wet days over the period, or approximately 22 additional rain days per year. This difference is statistically significant and operationally material because rain affects outdoor scheduling, guest movement, landscape maintenance, waterproofing needs, and the perceived quality of a nature-based retreat.

The seasonal control strengthens the interpretation. Both locations follow the same broad rainfall rhythm, with wetter autumn/winter periods and drier summers, but Vila Real is wetter in essentially every month and in all four seasons. Therefore, the precipitation difference is not a seasonal artefact created by one unusual part of the year; it is a structural difference between the two locations.

For atmosphere, the validation confirms that Guarda is consistently brighter and less cloudy, although the practical effect sizes are smaller than for rainfall. This means that atmospheric quality should not be used alone to justify the investment decision, but it should be treated as a reinforcing factor. In a premium wellness concept, even modest improvements in sunshine, clear-sky frequency, and lower summer cloud cover can strengthen the perceived quality of the guest experience when they align with stronger operational indicators.

Statistics, Stability, and Predictability

Overall Descriptive Statistics

Table 4.4 presents the mean, standard deviation, and coefficient of variation for the seven key meteorological variables across the full observation period.

Variable	Guarda			Vila Real		
	Mean	Std	CV (%)	Mean	Std	CV (%)
Max temperature (°C)	16.27	7.91	48.6	18.72	7.43	39.7
Min temperature (°C)	6.96	5.64	81.0	9.71	5.46	56.3
Mean temperature (°C)	11.41	6.73	59.0	13.96	6.28	45.0
Precipitation (mm/day)	2.31	5.78	249.9	3.93	8.88	225.9
Wind gusts (km/h)	39.52	13.11	33.2	39.30	14.27	36.3
Sunshine (hours/day)	9.12	3.87	42.4	8.82	4.10	46.5
Cloud cover (%)	51.19	33.78	66.0	54.10	34.30	63.4

Table 4.4: *Descriptive statistics for key meteorological variables Guarda and Vila Real, 2018early 2026.*

Several patterns stand out. Guarda consistently exhibits higher CVs for temperature variables than Vila Real, while Vila Real shows lower CVs for all temperature variables but a wetter baseline. For precipitation, both locations show very high CVs, which is expected given the skewness of daily rainfall data. Wind gust variability is similar in both locations.

Extreme Event Frequencies

Figure 4.18 presents the frequency of extreme days for each location, expressed as a percentage of all observation days.

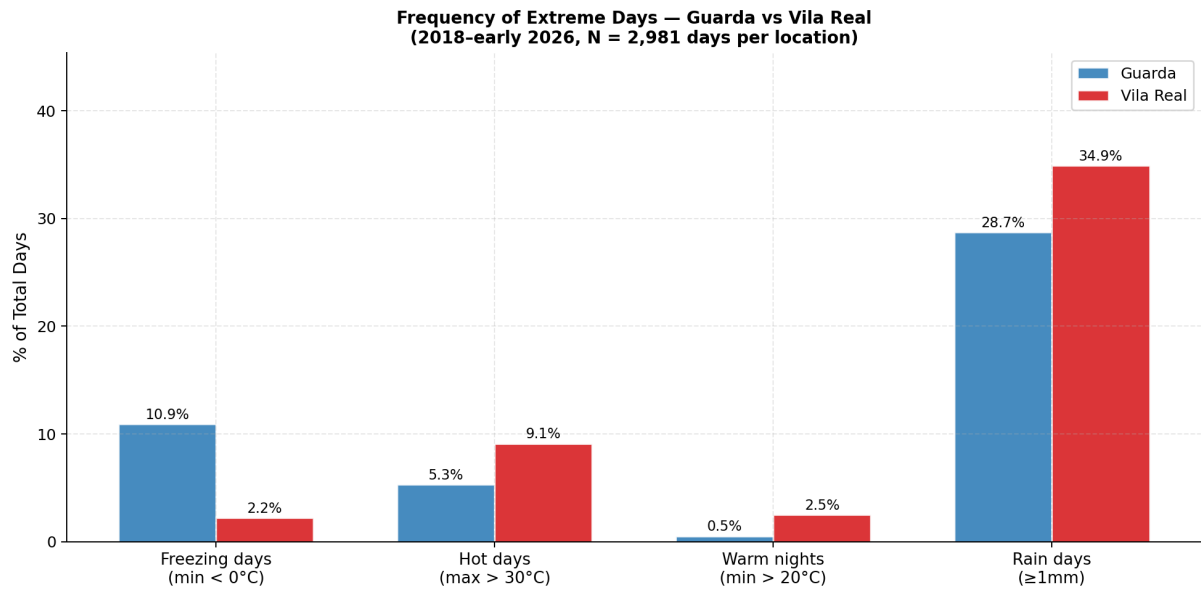


Figure 4.18: Frequency of extreme days (freezing, hot, warm night, rain) as a percentage of total days for Guarda and Vila Real.

The contrast between the two locations is pronounced and asymmetric. Guarda’s most salient extreme is freezing days, while Vila Real exhibits a much higher frequency of hot days, warm nights, and rain days.

Annual Consistency of Key Indicators

Indicator	Guarda			Vila Real		
	Mean/yr	Std	CV (%)	Mean/yr	Std	CV (%)
OK days	77.9	10.5	13.5	76.0	13.7	18.0
Freezing days	38.6	9.7	25.0	8.0	5.7	70.7
Hot days (>30°C)	19.9	11.8	59.5	33.8	16.3	48.3
Rain days	102.8	7.7	7.5	124.8	6.8	5.4
Precipitation (mm)	809	131	16.2	1372	135	9.8

Table 4.5: *Annual consistency of key indicators for Guarda and Vila Real (2018–2025).*

Table 4.5 presents the mean, standard deviation, and CV for key annual indicators, computed year by year from 2018 to 2025.

Figure 4.19 provides the same comparison visually. It highlights the central planning

trade-off: Guarda has higher cold exposure, but its OK-day supply is more stable year to year, while Vila Real combines higher rainfall burden with more variable OK-day availability.

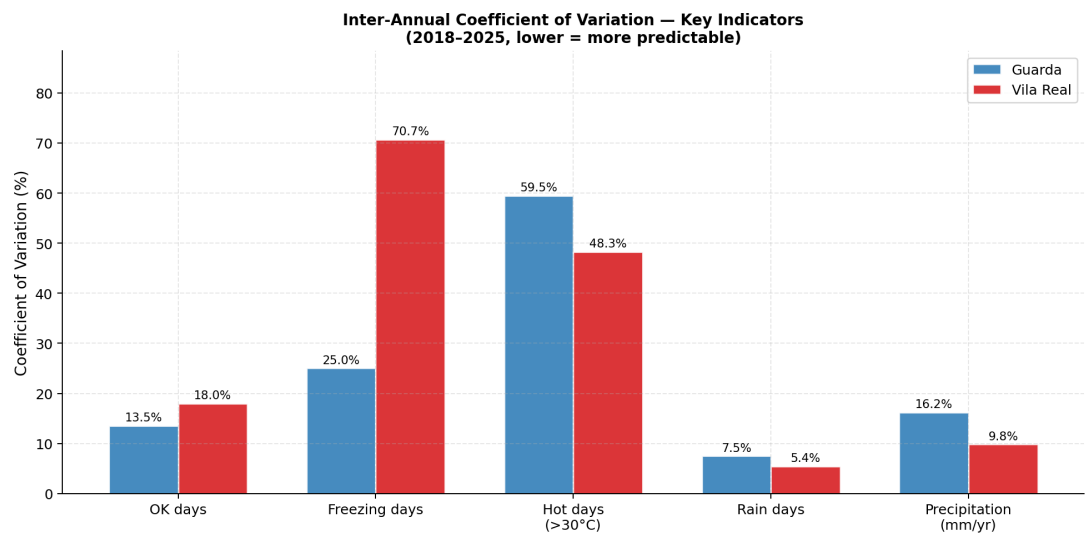


Figure 4.19: Annual coefficient of variation for operational climate indicators.

The results reveal a nuanced picture. For OK days, Guarda achieves a slightly higher mean and a notably lower CV. This means that not only does Guarda offer marginally more good outdoor days on average, but those days arrive more predictably from year to year.

Seasonal Analysis

Figure 4.20 presents the seasonal breakdown of mean maximum temperature, mean daily precipitation, and total OK days for each season and location.

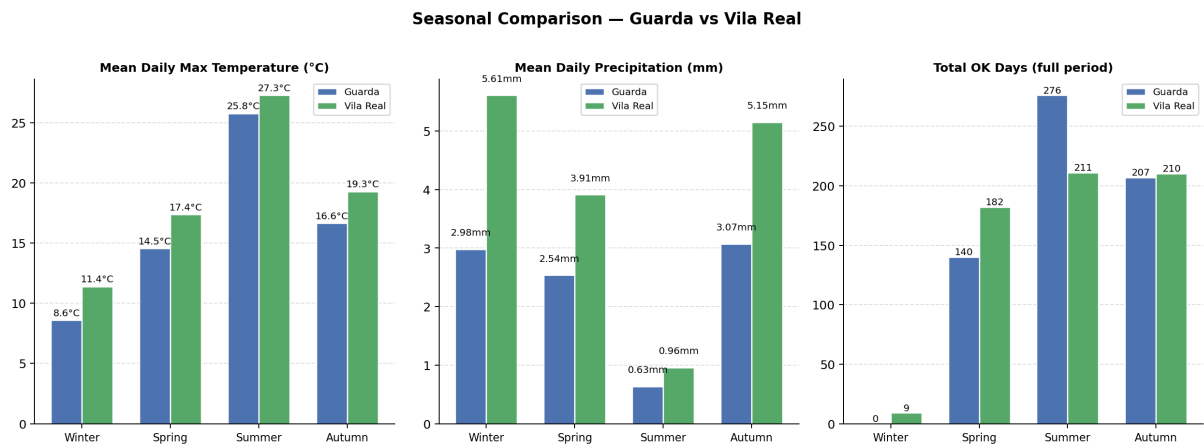


Figure 4.20: Seasonal comparison of mean maximum temperature, mean daily precipitation, and total OK days for Guarda and Vila Real.

The seasonal analysis reveals that the two locations trade advantages across the calendar year in ways that are directly relevant to the business model. In summer, Guarda performs most clearly better.

Summer Inter-Annual Consistency

The year-by-year analysis of summer maximum temperature shows that both locations are broadly stable across the observed years. However, summer precipitation is considerably more volatile. The key distinction is that Guarda’s lower absolute summer rainfall reduces the operational exposure.

Monthly Inter-Annual Temperature CV

Figure 4.21 presents the inter-annual CV of monthly mean temperature for each calendar month in both locations.

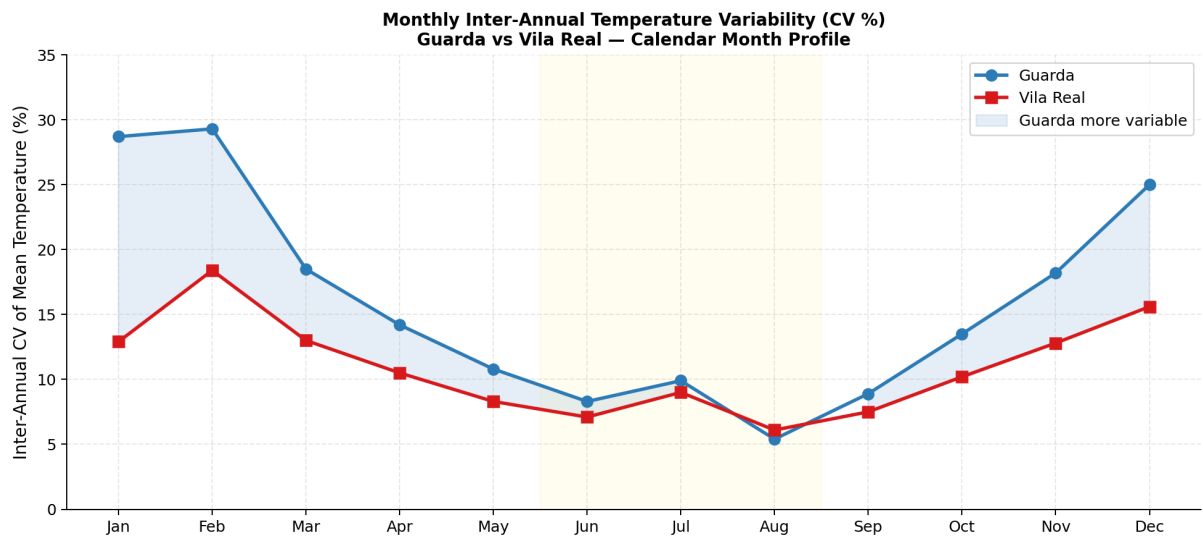


Figure 4.21: Inter-annual coefficient of variation (%) of monthly mean temperature for Guarda and Vila Real, by calendar month.

The pattern is consistent and informative. Guarda exhibits substantially higher inter-annual CV in every winter and shoulder month, while in summer the two locations converge.

Predictive Model Results OK Days

Table 4.6 summarises the planning-oriented OK-day prediction model. The target class shown is *OK*, because the practical business question is whether the model can identify

days suitable for outdoor wellness activities.

Table 4.6: Predictive model results for the OK class.

Model	AUC	CA	F1	Precision	Recall	MCC
Logistic Regression	0.868	0.816	0.475	0.582	0.401	0.377
Decision Tree	0.658	0.819	0.528	0.572	0.491	0.419
Random Forest	0.888	0.839	0.586	0.627	0.550	0.488
Naive Bayes	0.810	0.736	0.573	0.431	0.853	0.460

Figure 4.22 provides the model-performance comparison visually. It reinforces the table interpretation: tree-based and ensemble approaches capture the nonlinear relationship between seasonality, atmosphere, and OK-day probability better than a purely linear model. The model is not the basis of the recommendation by itself, but it confirms that OK days have a predictable structure that can support future scheduling.

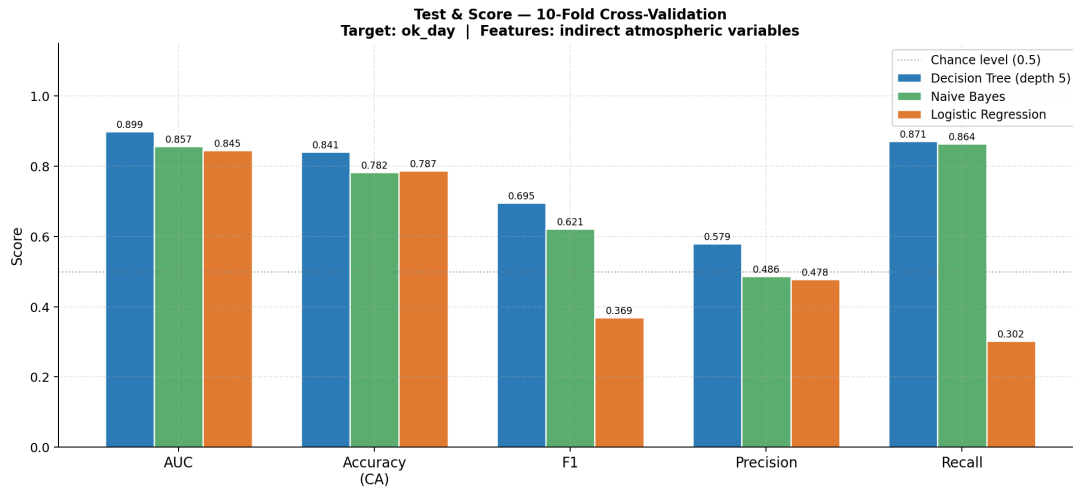


Figure 4.22: Classification model performance for OK-day prediction.

The Random Forest model currently provides the strongest overall balance, with the highest AUC (0.888), classification accuracy (0.839), F1-score for OK days (0.586), precision (0.627), and MCC (0.488). This suggests that nonlinear interactions between location, seasonality, and atmospheric variables are useful for anticipating suitable outdoor days.

Naive Bayes has a different profile. It achieves the highest recall for OK days (0.853), meaning that it captures most suitable days, but its lower precision (0.431) indicates a much higher number of false positives. In business terms, this model would be optimistic: it would identify many possible outdoor days, but it would also overstate suitability more

often. Random Forest is therefore more balanced for planning, while Naive Bayes may be useful if the priority is to avoid missing potential activity days.

Table 4.7 presents the confusion matrix for the Random Forest model. This model correctly identifies 679 of the 1,235 OK days and misclassifies 556 OK days as Not OK. It also produces 404 false positives, where the model predicts OK but the actual class is Not OK. This profile is reasonably conservative: it captures more than half of suitable outdoor days while avoiding the very high false-positive rate observed in the more optimistic model.

Table 4.7: Confusion matrix for the Random Forest OK-day model.

Actual / Predicted	Not OK	OK	Total
Not OK	4,323	404	4,727
OK	556	679	1,235
Total	4,879	1,083	5,962

For comparison, Naive Bayes correctly identifies 1,053 OK days but produces 1,389 false positives. This makes it useful when the priority is to avoid missing possible activity days, but less appropriate as the main planning model because it may overstate the number of days that are operationally suitable.

These results add value to the decision framework because they show that OK-day availability is not random: it is predictable from the location-calendar-atmosphere structure. This supports the use of OK days as an operational planning indicator rather than merely a descriptive count. The model should not replace the direct climatic comparison, but it confirms that the observed patterns have enough structure to support seasonal planning and operational forecasting.

Comparison Between Alternatives Stability and Predictability

The predictability dimension adds a critical layer to the location comparison that goes beyond average conditions. On the most operationally important indicator — annual OK days — Guarda is both higher in mean and lower in inter-annual CV.

The rainfall picture does not clearly favour either location on predictability grounds alone. What matters operationally is not which city is more predictably rainy, but which city offers more reliable conditions during the programming season. On that criterion, Guarda's

drier summers and lower summer rainfall burden provide a more operationally stable environment.

The statistical and predictability analysis indicates that Guarda has a stronger summer planning profile, while Vila Real has advantages in spring and winter mildness. Combined with the rainfall and atmosphere evidence, this strengthens the case for Guarda as the more appropriate location for a premium outdoor-oriented retreat.

Integrated Comparison Between Alternatives

This section consolidates the findings from all four analytical dimensions — comfort, risk and operations, predictability, and atmosphere — into a single structured comparison between Guarda and Vila Real.

Comfort

Both locations produce a nearly identical total number of OK days over the observation period. However, Guarda concentrates more of these days in summer, which is the aspect that matters most for a retreat business model.

Comfort interpretation: Guarda currently holds the advantage for the commercially critical summer season, while Vila Real remains stronger for spring and winter mildness.

Risk and Operations

The risk profile of the two locations is asymmetric. Vila Real’s primary operational risk is summer heat and rainfall burden, whereas Guarda’s primary operational risk is winter cold.

Table 4.8: Operational risk comparison: Guarda vs. Vila Real (2018–2026).

Risk indicator	Guarda	Vila Real	Season affected
Hot days ($>30^{\circ}\text{C}$)	159	270	Summer (peak season)
Warm nights ($>20^{\circ}\text{C}$)	15	75	Summer (peak season)
Freezing days ($<0^{\circ}\text{C}$)	326	67	Winter (low season)
Rain days ($\geq 1\text{ mm}$)	856	1039	Year-round

Risk interpretation: Guarda’s risks are concentrated mainly in the low season and

appear more infrastructure-manageable. Vila Real's risks are more exposed to the peak season, especially through heat and rain.

Predictability

On the most operationally critical indicator — annual OK days — Guarda is both marginally higher in mean and meaningfully more stable year to year.

Predictability interpretation: Guarda is currently more predictable on annual OK days, while Vila Real is more stable in rain-day counts relative to its wetter baseline.

Atmospheric Quality and Perceived Wellness Environment

On every atmospheric indicator examined, Guarda outperforms Vila Real.

Atmosphere interpretation: Guarda holds a clear and consistent advantage on the atmospheric dimension.

Integrated Trade-off Summary

Table 4.9 presents a structured summary of the comparative findings across all four dimensions.

Table 4.9: Integrated trade-off summary: Guarda vs. Vila Real across all four dimensions.

Dimension	Guarda	Vila Real	Advantage
Comfort (OK days, annual)	623 days	612 days	Tie
Comfort (OK days, summer)	276 days	211 days	Guarda
Comfort (OK days, spring)	140 days	182 days	Vila Real
Risk: heat (peak season)	159 hot days	270 hot days	Guarda
Risk: cold (low season)	326 freezing days	67 freezing days	Vila Real
Risk: rainfall (year-round)	856 rain days	1039 rain days	Guarda
Predictability (OK days CV)	13.5%	18.0%	Guarda
Predictability (rain days CV)	7.5%	5.4%	Vila Real
Atmosphere (sunshine/day)	9.12 h	8.82 h	Guarda
Atmosphere (cloud cover)	51.2%	54.1%	Guarda
Atmosphere (summer cloud)	28.0%	34.0%	Guarda

The trade-off structure is clear and consistent. Vila Real holds advantages in spring comfort, winter cold, and marginally in rain day predictability. Guarda holds advantages in summer comfort, peak-season heat risk, rainfall burden, annual OK day predictability, and atmospheric quality.

Decision Implications

The integrated analysis leads to the following interpretation:

- Guarda is stronger on summer usability, heat-risk control, rainfall burden, atmospheric quality, and annual OK-day stability.
- Vila Real is stronger on winter mildness, spring OK-day availability, and lower relative variation in rain-day counts.
- A summer-oriented outdoor retreat gives more weight to Guarda’s advantages, while a broader spring-and-winter calendar would give more relevance to Vila Real’s strengths.

Because the proposed concept is a premium mountain wellness retreat built around outdoor comfort, atmosphere, and reliable seasonal programming, the integrated evidence favours Guarda.

Table 4.10: Weighted decision matrix for the preferred location.

Criterion	Weight	Guarda score	Vila Real score	Weighted advantage
Summer outdoor usability	30%	9	6	Guarda
Rainfall disruption	20%	8	5	Guarda
Winter operation	15%	4	8	Vila Real
Atmospheric quality	15%	8	6	Guarda
Planning predictability	10%	7	6	Guarda
Strategic fit with concept	10%	8	6	Guarda
Weighted total	100%	7.45	6.05	Guarda

The weights in Table 4.10 make the decision logic explicit. They give the greatest importance to summer outdoor usability and rainfall disruption because these factors most directly affect the proposed retreat’s core activities and guest experience. The matrix does not hide Vila Real’s winter advantage, but shows that this advantage is not large enough to offset Guarda’s stronger performance on the higher-priority criteria.

Chapter 5

Limitations and Assumptions

5.1 Assumptions

The analysis relies on a set of explicit assumptions that should be kept visible when the results are interpreted.

First, the OK-day indicator is treated as a useful approximation of outdoor activity feasibility. A day classified as OK is assumed to be broadly suitable for activities such as hiking, outdoor yoga, meditation sessions, outdoor meals, and nature-based guest experiences. The thresholds used in this indicator are operational approximations rather than perfect measures of human comfort.

Second, summer and early autumn are assumed to have higher commercial relevance for the specific concept evaluated in this report. This assumption is based on the proposed product mix: outdoor yoga, hiking, meditation, terrace dining, scenic relaxation, and nature-based packages depend heavily on long daylight hours and guest willingness to spend time outside. Spring and winter can still support valuable programmes, especially indoor spa, recovery, and retreat formats, but they are treated as secondary because the analysed concept is not mainly an indoor winter resort. If the client later chooses a spring-oriented or winter-indoor positioning, the weighting of the decision matrix should be revised and Vila Real's advantages would become more important.

Third, the comparison between Guarda and Vila Real is assumed to be structurally comparable because both locations are analysed over the same period, with the same number of daily observations, the same variables, and the same derived indicators. This matters especially for the comparability discussion in the predictive model: **zone** is used as a comparison attribute, but the locations are business alternatives rather than protected human groups.

Fourth, the model assumes that historical patterns from 2018 to early 2026 provide a useful basis for medium-term planning. This is reasonable for comparative assessment, but it does not fully address long-term climate change, future extreme events, or changes in tourism demand.

Fifth, daily observations are treated as sufficiently independent for the comparative tests. In reality, weather conditions can be autocorrelated across consecutive days. This does not invalidate the descriptive comparison, but it means that formal significance levels should be interpreted alongside effect sizes, seasonal consistency, and operational relevance.

5.2 Limitations

The OK-day indicator simplifies a continuous comfort experience into a binary classification. A day classified as Not OK may still support some indoor or low-intensity activities, while a day classified as OK may not be ideal for every guest profile or every programme. This simplification is useful for modelling and comparison, but it should not be interpreted as a complete measure of guest experience.

The dataset is daily rather than hourly. This limits the ability to distinguish between rain or heat that occurs during activity hours and conditions that occur at less relevant times of day. For example, rainfall overnight may have a different operational impact than rainfall during a scheduled outdoor meditation session.

The model does not include financial variables such as heating costs, cooling costs, staffing costs, construction standards, occupancy rates, or expected revenue by season. For this reason, the analysis can identify climatic risk and operational implications, but it does not convert those risks into a complete investment model.

The statistical tests are influenced by the large number of daily observations. As a result, small differences can become statistically significant even when their practical effect is limited. This is why the report gives greater decision weight to variables where statistical evidence is paired with clear operational meaning, such as precipitation, summer OK days, heat exposure, and annual OK-day stability.

The dataset represents coordinate-level historical weather rather than full site-specific microclimate information. Local slope orientation, altitude variation, forest cover, wind exposure, building placement, and shade could modify the guest experience at the final property level.

The analysis excludes broader investment variables such as land cost, road access, construction permits, labour availability, competitive positioning, brand fit, and expected

seasonal revenue. The recommendation is therefore a climate-led location recommendation, not a complete financial due-diligence decision.

The predictive model is a supporting validation tool rather than a final decision engine. Its purpose is to confirm that OK-day availability has a predictable structure and can support future scheduling, not to replace the integrated business judgment.

Chapter 6

Final Recommendation

6.1 Recommendation to the Client

The recommended location for the Mountain Wellness retreat is Guarda.

The decision rule is specific: **choose Guarda if the resort is designed as a summer/autumn-led outdoor wellness retreat with strong winter indoor infrastructure. Choose Vila Real only if the client wants a milder, spring-oriented concept with lower winter operating risk and less emphasis on peak-season outdoor comfort.**

This recommendation is based on the alignment between Guarda's climatic profile and the client's intended value proposition: a premium mountain retreat centred on outdoor wellness, restorative atmosphere, scenic openness, and reliable seasonal programming. The annual number of OK days is similar in both locations, but the timing, risk structure, and operating implications of those days are not equivalent.

Guarda provides a stronger peak-season platform. It records 276 summer OK days compared with 211 in Vila Real, a difference of 65 usable summer days over the observation period. This matters because summer is the period in which outdoor yoga, hiking, meditation, terrace dining, and nature-based retreat packages are most commercially valuable. Vila Real's spring advantage is real, but it is less aligned with the core revenue season of a mountain wellness concept.

Guarda also carries a lower peak-season heat risk. Vila Real records 270 hot days above 30°C compared with 159 in Guarda, and 75 warm nights compared with only 15 in Guarda. These conditions affect guest comfort, sleep quality, cooling demand, and the feasibility of outdoor activities. By contrast, Guarda's main thermal disadvantage is winter cold, with 326 freezing days compared with 67 in Vila Real. This is important, but

it is more manageable through heating, insulation, indoor spa capacity, and low-season programming than peak-season heat and warm nights are for an outdoor retreat.

The precipitation evidence provides the strongest operational justification. Vila Real accumulated 11,714.1 mm of rainfall compared with 6,896.4 mm in Guarda, about 70% more rainfall. It also recorded 1,039 rain days compared with 856 in Guarda, or 183 additional wet days over the observation period. This gap is statistically robust ($t = 8.329$, $p < 0.001$) and practically meaningful because rainfall directly affects cancellations, guest movement, maintenance, humidity exposure, and the perceived quality of outdoor wellness experiences.

Guarda is also more reliable on the main planning indicator. Annual OK days have a lower coefficient of variation in Guarda than in Vila Real (13.5% vs. 18.0%), meaning that Guarda's usable outdoor calendar is more stable from year to year. The predictive modelling results further confirm that OK-day availability follows a structured pattern rather than random fluctuation, supporting the use of these indicators for planning and scheduling.

Finally, Guarda offers the stronger wellness atmosphere. It has higher mean daily sunshine (9.12 vs. 8.82 hours), lower mean cloud cover (51.2% vs. 54.1%), lower summer cloud cover (28.0% vs. 34.0%), more clear days, fewer overcast days, and fewer low-sunshine days. These differences are smaller than the rainfall difference, but they consistently reinforce Guarda's positioning as the brighter and more open-feeling retreat environment.

Why Vila Real Is Not the Preferred Option

Vila Real is not a weak alternative. It has milder winters, fewer freezing days, and more spring OK days. These advantages would become more important if the client wanted a retreat focused primarily on spring programming, shoulder-season mildness, or lower winter heating requirements.

However, Vila Real's disadvantages are concentrated in areas that matter more for this specific concept. It is hotter in summer, has more warm nights, is substantially wetter, and has a cloudier atmospheric profile. These risks affect the core guest experience and the highest-value outdoor programming season. The location therefore appears more exposed to operational disruption and comfort limitations during the period when the resort would most need to perform well.

Implementation Conditions for Guarda

The recommendation for Guarda should be implemented with three conditions:

- **Design for winter resilience:** invest in strong insulation, reliable heating, indoor spa capacity, and winter-specific wellness programming to manage Guarda’s colder low season.
- **Position the commercial calendar around summer and early autumn:** prioritise outdoor retreats, hiking, yoga, and open-air experiences during the months when Guarda’s OK-day advantage is strongest.
- **Complete financial and site due diligence:** before acquisition, validate land cost, road access, permitting, water availability, staffing, construction feasibility, and local microclimate at the exact property site.

With these conditions, Guarda offers the better balance of outdoor usability, atmospheric quality, operational reliability, and strategic fit for a premium mountain wellness retreat.

6.2 Next Steps

The recommendation is climate-led and should now be converted into an implementation decision through site-level due diligence. The next steps are:

1. validate the exact property microclimate, including elevation, slope orientation, wind exposure, shade, and local access conditions;
2. estimate the capital and operating cost of Guarda’s winter-resilience requirements, especially insulation, heating, indoor wellness capacity, and low-season guest amenities;
3. build a seasonal revenue plan that prioritises summer and early autumn outdoor packages, while using indoor spa, recovery, and retreat formats to reduce winter seasonality;
4. compare non-climatic investment constraints, including land cost, permits, infrastructure, staffing, supplier access, and competitive positioning;
5. update the climate risk assessment with future climate projections before committing to final acquisition and construction.

Appendix A

Reproducibility and Traceability

A.1 Tools Used

The analysis was prepared using a reproducible workflow combining Python, Orange Data Mining, and LaTeX. Python was used for data cleaning, derived-variable construction, aggregation, descriptive statistics, statistical validation, and visualisation. Orange Data Mining was used as a complementary visual modelling environment for classification model comparison and confusion-matrix validation. LaTeX was used to assemble the final written report. Spreadsheet-style tabular checks were used only for manual inspection of exported summaries and did not replace the scripted calculations.

A.2 Data Sources

The analysis uses two daily meteorological datasets:

- `raw_GUARDA.csv`: daily observations for Guarda;
- `raw_VILA_REAL.csv`: daily observations for Vila Real.

Both files cover the same period, contain the same variables, and contain 2,981 daily observations per location. This gives a combined analytical dataset of 5,962 observations.

A.3 Derived Variables

The following variables were derived from the raw files:

- `month`, `year`, and `season`, derived from the date field;
- `sunshine_hours`, calculated as `sunshine_duration/3600`;
- `rain_day`, equal to 1 when daily precipitation is at least 1.0 mm;
- `hot_day`, equal to 1 when daily maximum temperature exceeds 30°C;
- `freezing_day`, equal to 1 when daily minimum temperature is below 0°C;
- `warm_night`, equal to 1 when daily minimum temperature exceeds 20°C;
- `ok_day`, equal to 1 when precipitation is below 1.0 mm, wind gusts are below 40 km/h, and daily maximum temperature is between 18°C and 30°C.

A.4 Aggregation Logic

The descriptive tables were produced from the combined daily dataset using the following aggregation logic:

- annual comparisons group by `zone` and `year`;
- seasonal comparisons group by `zone` and `season`;
- monthly comparisons group by `zone` and `month`;
- binary indicators use sums to count days;
- continuous meteorological variables use means, standard deviations, and coefficients of variation where relevant.

A.5 Predictive Model Inputs

The predictive model uses the file `ok_day_model_simple.csv`. The target variable is `ok_day_label`, with two classes: *OK* and *Not OK*. The input features are:

- `zone`;
- `season`;
- `month`;
- `day_of_year`;

- `sunshine_hours`;
- `cloudcover_mean`;
- `relative_humidity_2m_mean`;
- `shortwave_radiation_sum`.

The direct rule variables `temperature_2m_max`, `precipitation_sum`, and `wind_gusts_10m_max` were excluded from the predictive model because they directly define the OK-day target.

A.6 Materials Checklist

- raw Guarda and Vila Real CSV files;
- combined OK-day modelling dataset;
- monthly and annual OK-day validation summaries;
- exported figures used in the report;
- final report PDF.

Appendix A

Reproducibility and Traceability

A.1 Tools Used

The precipitation analysis was carried out in Orange Data Mining. The statistical and predictability analysis was conducted using Orange Data Mining (version 3). All steps were performed through the graphical interface. The analysis was conducted using Orange Data Mining (version 3), a visual analytics platform. No programming was required. All steps were performed through the graphical interface using connected widgets.

A.2 How to Reproduce the Analysis

Precipitation Analysis

Input Data:

- date, zone, precipitation_sum, rain_day, month, season

General flow: File → Select Columns → Group By / Box Plot / Bar Plot / Distributions / Data Table

- Box Plot: precipitation_sum by zone.
- Group By (zone + month, Mean precipitation_sum) → Bar Plot.
- Group By (zone + season, Mean precipitation_sum) → Bar Plot.
- Group By (zone + month, Sum rain_day) → Bar Plot.

- Box Plot: monthly precipitation_sum grouped by month (separately for each city).

Export and Verification: Graphs were exported as image files. Summary values were verified through the Data Table output after aggregation.

Statistical and Predictability Analysis

Step 1 — Load data and create all derived variables

Follow Steps 1–4 of Appendix A (Comfort Indicators) to load, concatenate, and create month, year, season, sunshine_hours, and all extreme day indicators (freezing_day, hot_day, warm_night, rain_day, ok_day).

Step 2 — Group By: overall statistics by zone

- Drag Group By. Connect from the full prepared dataset.
- Group by zone (2 instances). Aggregate Mean for all meteorological variables and Mean + Sum for all binary indicators.
- Rename: agrupar por zone.

Step 3 — Group By: annual aggregation

- Drag Group By. Group by zone and year (18 instances: 9 years $\times$ 2 cities).
- Aggregate Sum for ok_day, freezing_day, hot_day, rain_day, precipitation_sum; Mean for temperature_2m_max.
- Rename: agrupar por zone e year.
- Connect to Bar Plot widgets for annual OK days, freezing days, hot days.

Step 4 — Group By: seasonal aggregation

- Drag Group By. Group by zone and season (8 instances).
- Aggregate Mean for temperature_2m_max and precipitation_sum; Sum for ok_day.
- Rename: agrupar por zone e season.
- Connect to three Bar Plot widgets: seasonal temperature, seasonal precipitation, seasonal OK days.

Step 5 — Group By: monthly aggregation

- Drag Group By. Group by zone and month (24 instances).
- Aggregate Mean for temperature_2m_mean.
- Rename: agrupar por zone e month.
- Connect to Bar Plot: monthly mean temperature.

Step 6 — Box Plots for extreme indicators

- Box Plot: temperature_2m_max by zone → save boxplot_temp_max.png.
- Box Plot: precipitation_sum by zone → save boxplot_precip.png.

Step 7 — Define ok_day as target (Edit Domain)

- Drag Edit Domain. Connect from the full prepared dataset.
- Locate ok_day. Change its Role to Target. Click Apply.
- Rename: definir ok_day como target.

Step 8 — Select predictor features

- Drag Select Columns. Move all to Ignored, then move back as Features: temperature_2m_mean, temperature_2m_min, sunshine_hours, cloudcover_mean, relative_humidity_2m_mean, shortwave_radiation_sum, month.
- Confirm ok_day appears as Target. Rename: seleccionar features predictoras.

Step 9 — Add learners and run Test & Score

- Add Tree (max depth 5), Logistic Regression, and Naive Bayes from the Model tab.
- Drag Test and Score. Connect the feature-selected data and all three learners.
- Set Cross validation, 10 folds. Save results screenshot as test_and_score_results.png.
- Add Confusion Matrix (connect Evaluation Results, select Tree model) → save confusion_matrix_tree.png.
- Add ROC Analysis (connect Evaluation Results) → save roc_analysis.png.

Integrated Comparison

Purpose: This section documents how to reproduce the integrated comparison presented in Section 4.2 using Orange Data Mining. Unlike the thematic appendices, which each cover a single analytical dimension, this section describes how to bring all previously computed indicators together into a single consolidated view and how to produce the two summary tables (Table 4.2 and Table 4.3) that support the integrated trade-off analysis and the final recommendation.

No new raw data processing is required. All the variables and aggregations needed for this section are built on top of the datasets and workflows already described in the preceding appendices. The steps below assume that the combined and feature-enriched dataset (containing `ok_day`, `freezing_day`, `hot_day`, `warm_night`, `rain_day`, `sunshine_hours`, `cloudcover_mean`, `month`, `year`, and `season`) is already available in Orange, as constructed in Appendix A (Comfort Indicators).

Step 1 — Load the enriched dataset

- Open the Orange workflow from Appendix A (Comfort Indicators) which already contains the full concatenated and feature-enriched dataset at the output of the widget `criar indicadores`.
- If starting from scratch, follow Steps 1–4 of Appendix A to load `raw_GUARDA.csv` and `raw_VILA_REAL.csv`, concatenate them, and create all derived variables (`month`, `year`, `season`, `sunshine_hours`, `ok_day`, `freezing_day`, `hot_day`, `warm_night`, `rain_day`).
- Confirm the dataset shows 5962 instances before proceeding.

Step 2 — Group By: aggregate all key indicators by zone

- From the Transform tab, drag a Group By widget onto the canvas.
- Connect the output of `criar indicadores` to this Group By.
- Double-click to open.
- In the left panel, click `zone` to select it as the only grouping variable. Confirm the bottom left corner shows 2 instances (`GUARDA` and `VILA_REAL`).
- In the right panel, set the following aggregations: `ok_day` → Sum; `freezing_day` → Sum; `hot_day` → Sum; `warm_night` → Sum; `rain_day` → Sum; `precipitation_sum` → Mean; `sunshine_hours` → Mean; `cloudcover_mean` → Mean.

- Confirm Send Automatically is checked. Close and rename the widget to totais por zone.

Step 3 — Data Table: inspect and record the overall totals

- From the Data tab, drag a Data Table widget.
- Connect the output of totais por zone to it.
- Double-click and read off the values for GUARDA and VILA_REAL across all aggregated columns. These figures correspond directly to the rows of Table 4.2 (Operational Risk Comparison) in the report.
- Save a screenshot as datatable_totais_zone.png.

Step 4 — Group By: aggregate OK days by zone and season

- From the Transform tab, drag a new Group By widget.
- Connect the output of criar indicadores to it.
- Double-click. In the left panel, select zone and season as grouping variables. Confirm 8 instances (4 seasons $\times$ 2 cities).
- Set aggregation for ok_day to Sum.
- Confirm Send Automatically. Close and rename to okdays por zone e season.

Step 5 — Bar Plot: OK days by season and city

- From the Visualize tab, drag a Bar Plot widget.
- Connect the output of okdays por zone e season to it.
- Double-click to open.
- Under Values select ok_day – Sum.
- Under Group by select season.
- Under Color select zone.
- Save the image as barplot_okdays_seasonal_comparison.png.

Step 6 — Group By: aggregate all indicators by zone and year

- From the Transform tab, drag a new Group By widget.
- Connect the output of criar indicadores to it.
- Double-click. Select zone and year as grouping variables. Confirm 18 instances (9 years $\times$ 2 cities).
- Set the following aggregations: ok_day $\rightarrow$ Sum; freezing_day $\rightarrow$ Sum; hot_day $\rightarrow$ Sum; rain_day $\rightarrow$ Sum; sunshine_hours $\rightarrow$ Mean; cloudcover_mean $\rightarrow$ Mean.
- Confirm Send Automatically. Close and rename to indicadores por zone e year.

Step 7 — Data Table: inspect annual values for CV calculation

- From the Data tab, drag a Data Table widget.
- Connect the output of indicadores por zone e year to it.
- Double-click and filter rows to show only GUARDA or only VILA_REAL using the search bar at the top.
- Note the year-by-year values for ok_day – Sum for each city. These values allow the computation of the inter-annual mean, standard deviation, and coefficient of variation (CV) reported in Table 4.3 (Integrated Trade-off Summary).
- Save a screenshot as datatable_annual_indicators.png.

Step 8 — Bar Plot: multi-indicator comparison by zone

- From the Visualize tab, drag a Bar Plot widget.
- Connect the output of totais por zone to it.
- Double-click to open.
- Under Values select ok_day – Sum.
- Under Group by select zone.
- Under Color select zone.
- Save the image as barplot_okdays_total.png.
- Repeat for hot_day – Sum, freezing_day – Sum, and rain_day – Sum, saving each as barplot_hotdays_total.png, barplot_freezingdays_total.png, and barplot_raindays_total.png respectively.

Step 9 — Box Plot: sunshine hours and cloud cover by city

- From the Visualize tab, drag a Box Plot widget.
- Connect the output of criar indicadores (the full dataset) to it.
- Double-click. Under Variable select sunshine_hours. Under Subgroups select zone.
- Save the image as boxplot_sunshine_comparison.png.
- Repeat with a second Box Plot widget, this time selecting cloudcover_mean as the variable. Save as boxplot_cloudcover_comparison.png.

How the Summary Tables Were Constructed:

Table 4.2 (Operational Risk Comparison) was built directly from the output of totais por zone (Step 2 above). The Sum values for ok_day, hot_day, warm_night, freezing_day, and rain_day for each city correspond exactly to the figures reported in that table.

Table 4.3 (Integrated Trade-off Summary) was built by combining:

- Overall totals from totais por zone (Steps 2–3);
- Seasonal OK day totals from okdays por zone e season (Steps 4–5);
- Annual mean and CV values computed from the year-by-year output of indicadores por zone e year (Steps 6–7);
- Atmospheric means (sunshine_hours and cloudcover_mean) from totais por zone.

The coefficient of variation (CV) values in Table 4.3 were computed manually from the Data Table output: $CV = (\text{standard deviation} / \text{mean}) \times 100$. Orange does not compute CV directly, so the year-by-year values were read from the Data Table and the calculation was performed outside Orange.

A.3 Submitted Materials Checklist

Precipitation Analysis:

- Original CSV dataset and Orange workflow file (.ows)
- precipitationbyzone_boxplot.png, precipitationsumbymonth.png
- rainbyseason_barplot.png, raindaysbymonth.png

- guarda_boxplot.png, vilareal_boxplot.png

Statistical and Predictability Analysis:

- Report PDF and Orange workflow file (.ows)
- barplot_okdays_anual.png, barplot_freezingdays_anual.png, barplot_hotdays_anual.png
- barplot_temp_seasonal.png, barplot_precip_seasonal.png, barplot_okdays_seasonal.png
- barplot_temp_mensal.png, boxplot_temp_max.png, boxplot_precip.png
- test_and_score_results.png, confusion_matrix_tree.png, roc_analysis.png

Integrated Comparison:

- Report PDF
- raw_GUARDA.csv — raw meteorological data for Guarda
- raw_VILA_REAL.csv — raw meteorological data for Vila Real
- datatable_totais_zone.png
- datatable_annual_indicators.png
- barplot_okdays_seasonal_comparison.png
- barplot_okdays_total.png
- barplot_hotdays_total.png
- barplot_freezingdays_total.png
- barplot_raindays_total.png
- boxplot_sunshine_comparison.png
- boxplot_cloudcover_comparison.png
- Orange workflow file (.ows)